Trade Shocks and Media Slant*

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Abstract

Does trade liberalization reshape the political language of local news? I scale over 52 million newspaper articles against the evolving political vocabulary of the U.S. House of Representatives and use an event study exploiting variation in NAFTA exposure across commuting zones. Trade-exposed newspapers underwent a systematic rightward shift: the share of Republican-leaning articles increased by 9.6 percentage points for a top-versus-bottom-quartile difference in trade vulnerability. The shift is broad—not driven by trade-specific language—and appears across nearly all article categories. It emerges after NAFTA, persists through the sample period, and concentrates in low-education communities. Initially Democratic-leaning newspapers account for the entire shift, suggesting a demand-side channel.

JEL Codes: D72, F13, L82
Keywords: Media slant, polarization, NAFTA, local media, text-as-data

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1 Introduction

Trade shocks reshape not only local labor markets but also the political behavior of affected communities (Autor et al., 2013, 2020; Choi et al., 2024). Media slant, in turn, shapes voter beliefs and electoral outcomes (DellaVigna and Kaplan, 2007; Martin and Yurukoglu, 2017). Yet whether trade shocks also reshape the political language of local news—a potentially important link between economic dislocation and political change—is unknown.

This paper estimates the causal effect of trade exposure on the political language of local media. I measure media slant by scaling over 52 million local news articles against the evolving partisan vocabulary of the U.S. House of Representatives, building on Gentzkow and Shapiro (2010) and Widmer et al. (2025). Each article receives a predicted probability that its language resembles Republican rather than Democratic congressional speech; I classify it as Republican-leaning if this probability exceeds 0.5 and aggregate the share of R-leaning articles to the newspaper–year level. Using an event study that exploits variation in NAFTA exposure across Commuting Zones (Hakobyan and McLaren, 2016), I find that a top-versus-bottom-quartile shift in NAFTA vulnerability increases the share of Republican-leaning articles by 9.6 percentage points ($p < 0.01$; permutation $p < 0.001$).

Crucially, the shift is broad rather than mechanical: trade-specific language accounts for a negligible share of the partisan vocabulary, and zeroing all trade-related coefficients leaves the estimate virtually unchanged. The effect emerges after NAFTA and persists through the sample period, concentrates in low-education communities, and reflects convergence—initially Democratic-leaning newspapers drive the entire response.

NAFTA was the first major trade shock of the modern era, taking effect in 1994—when local newspapers were still the dominant source of political information and the internet had not yet disrupted local news markets. If trade shocks reshape local media language, the effect is most likely to emerge—and matter—when newspapers still hold a near-monopoly on local political information.

The results contribute to two literatures. First, while the existing literature emphasizes

demand-side factors (reader ideology) and supply-side factors (ownership, competition) as determinants of media slant (Gentzkow and Shapiro, 2010; Mullainathan and Shleifer, 2005; Gentzkow et al., 2011), this paper documents that real economic shocks can shift the partisan language of local news, with convergence evidence pointing to demand-side reader preferences as the most likely channel. The shift is not driven by trade-specific vocabulary: it appears across virtually all article categories, as confirmed by a topic-level decomposition. Second, the results complement the growing evidence that trade liberalization reshapes political behavior (Autor et al., 2020; Choi et al., 2024) by documenting that a key intermediary—the local newspaper—responds to the same shock, potentially amplifying the political consequences of economic dislocation.

The convergence pattern points to a demand-side channel: the shift is driven entirely by initially Democratic-leaning newspapers, consistent with newspapers adjusting content to match the changing preferences of readers who turned away from the party that signed NAFTA (Choi et al., 2024). A pure supply-side story—in which the loss of local advertising clients forces content changes—cannot explain why the effect depends on a newspaper's initial partisan lean.

2 Related Literature

Determinants of media slant. What drives the partisan positioning of news outlets? Mullainathan and Shleifer (2005) model how competition between outlets generates partisan differentiation when readers prefer like-minded news. Gentzkow and Shapiro (2010) develop a method for measuring newspaper slant using congressional speech patterns and find that demand-side factors—particularly the political composition of a newspaper's readership—explain far more of the variation in slant than supply-side factors such as ownership. Gentzkow and Shapiro (2006) show how reputational incentives can discipline or distort media reporting depending on the information environment. Other work examines how advertising revenues

(Petrova, 2012), ownership (Gentzkow and Shapiro, 2010), and geographic peer effects among outlets (Widmer et al., 2025) shape media positioning. This paper documents that real economic shocks can also shift media language, though the underlying channel remains an open question.

Consequences of trade liberalization. While the early trade-and-labor literature focused on wages and employment (Autor et al., 2013; Hakobyan and McLaren, 2016; Trefler, 2004), recent work has documented far-reaching effects on public goods provision (Feler and Senses, 2017), mortality (Pierce and Schott, 2020), crime (Dix-Carneiro et al., 2018), and political realignment (Autor et al., 2020; Choi et al., 2024). Choi et al. (2024) provide the closest precedent for this paper's empirical setting, documenting that NAFTA-induced trade exposure drove affected counties toward the Republican Party. Most directly related is Lu et al. (2018), who study the effect of Chinese import competition on 147 U.S. newspapers over 1998–2012 using the Autor et al. (2013) trade shock. Lu measures how trade exposure changes the *tone* of China-related coverage—finding that exposed newspapers report more negatively about China and are more likely to endorse Democrats. My paper differs in the outcome, the shock, and the setting. On the outcome: I ask whether trade exposure shifts the *overall partisan language* of a newspaper, not just coverage of a specific country. On the shock and setting: NAFTA took effect in 1994, when local newspapers were the dominant source of political information and the internet had not yet disrupted local news markets; the Chinese import surge accelerated after WTO accession in 2001, by which point online news was beginning to erode the newspaper's role as local information gatekeeper.

Text as data in political economy. This paper builds on the congressional-speech scaling approach of Gentzkow and Shapiro (2010) and the high-dimensional text analysis framework of Gentzkow et al. (2019). Related text-scaling methods include Laver et al. (2003), Monroe et al. (2008), and Taddy (2013). I adopt key design choices from Widmer et al. (2025)—training on all Republican and Democratic speeches and restricting features

to a shared congressional–newspaper vocabulary—but employ L1-penalized logistic regression rather than the multinomial specification of Widmer et al. (2025), with regularization chosen by 5-fold cross-validation. This allows the partisan vocabulary to evolve over time while maintaining cross-year comparability.

3 Institutional Background

NAFTA and differential trade exposure. The North American Free Trade Agreement, negotiated under President George H.W. Bush and signed into law by President Clinton in December 1993, took effect on January 1, 1994. The agreement eliminated most tariffs on goods traded among the United States, Canada, and Mexico, phasing in reductions over 5 to 15 years depending on the product, with the largest eventual reductions in manufacturing sectors where tariffs were high and Mexico held a comparative advantage, such as textiles, apparel, and footwear (Hakobyan and McLaren, 2016). Clinton's promotion of NAFTA alienated parts of the Democratic base in trade-exposed communities; Choi et al. (2024) document that affected counties subsequently shifted toward the Republican Party, consistent with voters turning away from the party that championed the agreement. Because pre-NAFTA tariff rates varied widely across industries—from near zero on many intermediate goods to over 30% on selected consumer products—and because industries are geographically concentrated, the agreement generated substantial cross-regional variation in trade exposure. Communities in the Midwest and Southeast whose employment was concentrated in high-tariff industries with Mexican comparative advantage experienced the sharpest adjustment, while service-oriented metropolitan areas were largely unaffected.

Local newspaper markets in the 1990s. During the sample period (1987–2004), newspapers remained the dominant source of local news in the United States. Most local markets were served by a single daily newspaper or, in larger cities, by two competing dailies (Gentzkow et al., 2011). Revenue derived primarily from advertising, supplemented by sub-

scription and newsstand sales, giving newspapers a strong incentive to maximize readership within their geographic market. A small local editorial staff made decisions about story selection and framing, typically responding to reader preferences and advertiser interests. The internet had not yet disrupted local news markets: as of 2000, fewer than half of U.S. adults used the internet regularly, and online news consumption remained marginal. This setting makes local newspapers a natural lens for studying how economic shocks shape the political language of local media, because both the shock (trade exposure) and the medium (the local newspaper) vary at the same geographic level.

4 Data and Measurement

4.1 Data Sources

I combine four main data sources. Newspaper articles (1987–2004) come from the NewsLibrary archive, linked to county FIPS codes by extracting city and state names from publication titles and matching them to county identifiers. I filter out non-news content (obituaries, classifieds, television listings, and other non-editorial items) using title-based keyword patterns. Congressional speeches come from the Hein-Bound corpus (Gentzkow et al., 2019) for House members in Congresses 99–108, merged with Nokken–Poole ideology scores from Voteview (Lewis et al., 2024). Trade and employment data come from Feenstra et al. (2002), the USITC, Hakobyan and McLaren (2016), and County Business Patterns. Editor & Publisher Yearbook data (ICPSR 30261; Gentzkow et al., 2011) provide newspaper-level advertising rates, circulation, and ownership at five snapshot years (1988, 1992, 1996, 2000, 2004); I match 128 of the 131 estimation-sample newspapers to this database using county FIPS codes and fuzzy name matching (98% match rate). Appendix A provides full details on data construction, sample restrictions, and the article filtering pipeline. Table 6 documents the sample construction from the initial NewsLibrary universe to the final panel of 131 daily newspapers in 98 commuting zones.

4.2 Measuring Trade Exposure

To measure local exposure to trade liberalization, I exploit the differential impact of NAFTA across U.S. regions based on their initial industrial composition. For each SIC industry, I compute an import-weighted ad valorem equivalent (AVE) tariff on Mexican imports and Mexico's Revealed Comparative Advantage ($RCA_{i,\text{Mex}}$), defined as the ratio of Mexico's export share in industry i to the rest-of-world average. Appendix B details the construction of AVE tariffs from HS8-level data and the concordances used.

The vulnerability of county c is defined as:

$$Vul_c = \frac{\sum_i L_{ci,1990} \times RCA_{i,\text{Mex}} \times \bar{\tau}_i}{\sum_i L_{ci,1990} \times RCA_{i,\text{Mex}}} \tag{1}$$

where $L_{ci,1990}$ is employment in industry i in county c in 1990, $\bar{\tau}_i$ is the import-weighted AVE tariff on Mexican imports in industry i, and $RCA_{i,\text{Mex}}$ is Mexico's revealed comparative advantage. The measure is an employment-and-RCA-weighted average tariff: counties whose workers are concentrated in industries where Mexico has comparative advantage and where tariffs are high face greater exposure to NAFTA liberalization.

I evaluate vulnerability at its 1990 level and scale it by the difference in mean vulnerability between the top and bottom quartiles of the cross-county distribution. This scaling ensures that one unit of the vulnerability measure corresponds to the gap between the average bottom-quartile and the average top-quartile county. The construction follows Choi et al. (2024), who build on Hakobyan and McLaren (2016). County-level variables are collapsed to the commuting zone (CZ) level (Tolbert and Sizer, 1996) using 1990 population weights.

Figure 1 maps the geographic distribution of NAFTA vulnerability across commuting zones. Exposure is concentrated in the manufacturing belt of the Midwest and parts of the Southeast, reflecting these regions' dependence on industries where Mexico holds a comparative advantage. The 98 commuting zones in the estimation sample cover the full range of the national vulnerability distribution (Figures 8 and 9 in Appendix E).

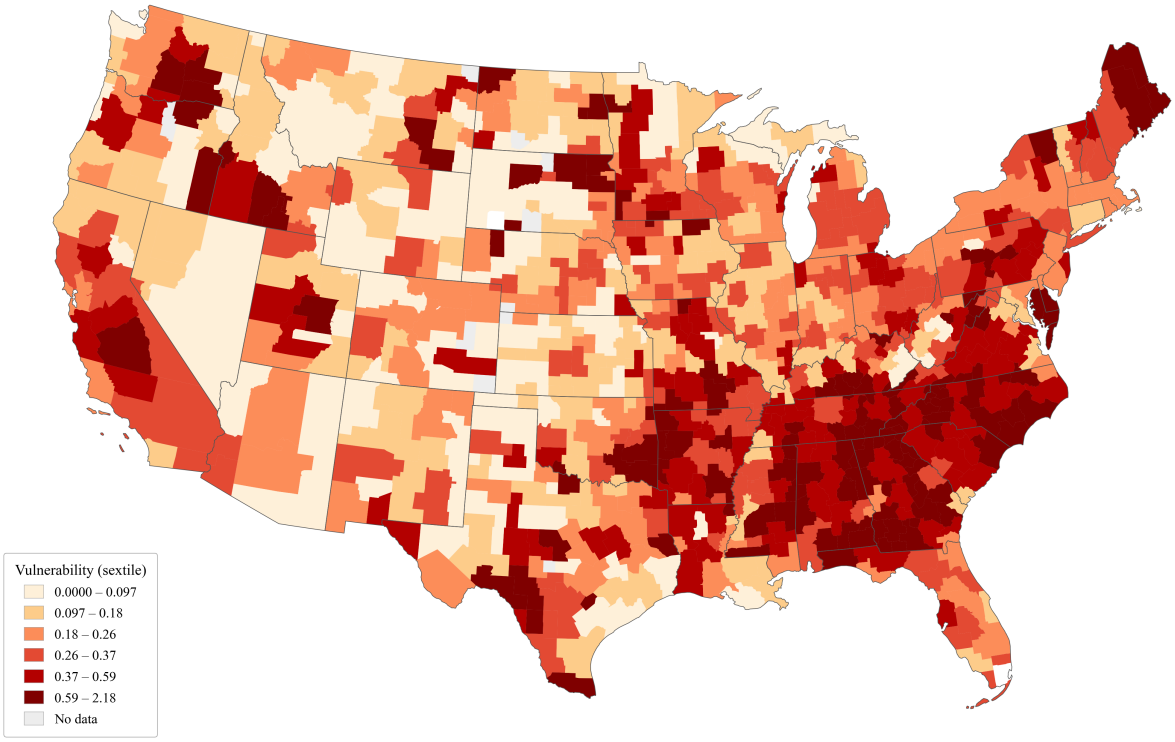

Figure 1: NAFTA Tariff Vulnerability by Commuting Zone (1990)

Notes: The map shows the 1990 NAFTA vulnerability index at the commuting zone level, defined as the employment-and-RCA-weighted average tariff on Mexican imports. Colors correspond to sextile bins computed over all 713 commuting zones, so that each color band contains an equal number of CZs. Gray areas indicate missing data. Figure 9 in Appendix E shows the 98 sample commuting zones.

A potential concern with the vulnerability measure is that it may correlate with pre-existing differences in media language or local characteristics, threatening the parallel trends assumption. Table 1 addresses this by comparing pre-NAFTA (1987–1993) means of key variables across commuting zones above and below the median vulnerability. Pre-treatment Share R-leaning is statistically indistinguishable between high- and low-vulnerability areas, confirming that trade-exposed and unexposed areas had similar political language before NAFTA. High-vulnerability CZs do have larger manufacturing sectors, lower education levels, and lower incomes—differences that arise mechanically from the construction of the vulnerability index, which weights by manufacturing employment. The division-by-year fixed

effects and the manufacturing share and China shock controls in the regression specification (Section 5) absorb these socioeconomic differences. Trade exposure is thus orthogonal to pre-treatment media language, supporting a causal interpretation of the post-NAFTA media response.

Table 1: Balance Test: Pre-NAFTA Characteristics by Vulnerability

	Low vulnerability ($N = 49$)	High vulnerability ($N = 49$)	Difference	p-value
Panel A: Pre-NAFTA media outcome (1987–1993)				
Share R-leaning, 1987–1993	0.439	0.430	-0.009	0.34
Panel B: Socioeconomic characteristics				
Manuf. empl. share, 1990	0.193	0.230	+0.037	0.02
Share less than high school, 1990	0.210	0.258	+0.047	<0.01
Share bachelor's or higher, 1990	0.221	0.184	-0.037	<0.01
Per capita income, 1989 ($1000s)	44.43	40.30	-4.13	<0.01
Population, 1990 (1000s)	1,636	1,406	-230	0.59
China import shock	0.767	1.26	+0.497	<0.01

Notes: Commuting zones split at the median of NAFTA vulnerability (1990). Panel A reports the pre-NAFTA (1987–1993) mean of the main outcome variable, averaged across newspaper–years within each CZ. Panel B reports 1990 socioeconomic characteristics. The "Difference" column reports High − Low. p-values are from two-sided Welch t-tests.

4.3 Measuring Media Slant

Following Gentzkow and Shapiro (2010), I measure media slant by projecting newspaper text onto the partisan dimension of congressional rhetoric. The approach first learns which phrases distinguish Republican from Democratic legislators, then measures how closely each newspaper article's vocabulary resembles one party's rhetoric versus the other's.

4.3.1 Congressional Speech Data

I combine House floor speeches from the Hein-Bound corpus (Gentzkow et al., 2019) (Congresses 99–108) with legislator ideology and party affiliation data from Voteview (Lewis et al., 2024). I match Hein-Bound speakers to Voteview records using normalized last names, state, and congress, disambiguating ties via first-name token overlap. The baseline specification

trains on *all* Republican and Democratic speeches (53,000–82,000 total per Congress), following Widmer et al. (2025). As a robustness check, I also train on a partisan core restricted to the top and bottom 20th percentiles of the Nokken–Poole ideology distribution (Section 6.4).

4.3.2 Text Preprocessing and Feature Selection

I preprocess congressional speeches to isolate ideological content from procedural, geographic, and personal-name language. The pipeline removes an expanded set of parliamentary formulae and procedural stop words following Gentzkow et al. (2019), geographic terms (state and city names), and legislator name bigrams (including nicknames), then stems all remaining tokens using the Porter stemmer. Distinctive legislator surnames that are not common English words are also removed as stop words. The final feature set consists of stemmed *unigrams and bigrams*. Including unigrams alongside bigrams expands the partisan vocabulary to capture individual words with strong partisan connotations (e.g., "wealthiest," "bureaucraci," "unborn") while bigrams preserve contextual meaning (e.g., "tax relief" vs. "tax break"). Each individual speech serves as one document. I apply a relative-frequency vectorizer (count-based with L1 row normalization) and a per-party frequency filter: a feature is retained if it appears in at least 0.1% of either party's speeches *and* in at least 0.01% of both parties' speeches. Finally, I keep only terms that also appear in at least 0.001% of newspaper articles, so that the LASSO trains exclusively on phrases it can later observe in newspaper text. This *shared vocabulary* restriction automatically excludes congressional jargon—such as procedural phrases and chamber-specific terminology—that never appears in newspaper articles. Appendix C provides the full list of filtering rules and preprocessing steps.

4.3.3 Training the Slant Model

To capture the evolving nature of political language, I train the slant model using a rolling window approach. For each Congress $t \in \{100, \ldots, 108\}$, I train a separate logistic regression

model on all individual speeches from Congress t, yielding nine models. The model predicts whether a speech was delivered by a Republican or Democratic legislator. I use LASSO (Tibshirani, 1996) (L1) regularization to select the most predictive features while driving irrelevant coefficients to zero, choosing the regularization strength by 5-fold cross-validation over a grid of 20 candidate values.[1]

Let y_i be an indicator for whether speech i was delivered by a Republican. The probability is modeled as:

$$P(y_i = 1 \mid \mathbf{x}_i) = \frac{\exp(\alpha_t + \mathbf{x}_i'\boldsymbol{\beta}_t)}{1 + \exp(\alpha_t + \mathbf{x}_i'\boldsymbol{\beta}_t)} \tag{2}$$

where $\mathbf{x}_i$ is the relative-frequency feature vector of speech i.

Training accuracy ranges from 68% to 76% at the speech level, and the model selects between 3,000 and 5,000 non-zero features per Congress (Table 11 in Appendix F). These rates reflect the inherent difficulty of classifying individual floor speeches, most of which are not overtly partisan. However, because each legislator delivers dozens to hundreds of speeches per Congress, aggregation sharply reduces noise: accuracy exceeds 90% at 10 speeches and 97% at 50 (Appendix D). The same convergence logic applies to newspapers, where each newspaper–year averages over thousands of articles. The shared vocabulary filter ensures that these features are interpretable and observable in newspaper text.

4.3.4 Scoring Newspapers

I apply the pre-trained vectorizer to newspaper articles, transforming each article into the same feature space as the congressional speeches. For each article d published during Congress t, I compute the predicted probability that its language resembles Republican

[1] This differs from Widmer et al. (2025), who models word frequencies with a multinomial distribution approximated via Poisson regression. I instead treat partisan classification as a binary prediction problem and use a standard logistic regression with L1 penalty (LASSO), which directly selects a sparse set of discriminating features. The two approaches share the same goal—identifying partisan language from congressional speeches—but differ in the statistical model.

rather than Democratic speech:

$$\hat{P}(\mathrm{R} \mid d) = \frac{\exp(\hat{\alpha}_t + \mathbf{x}_d' \hat{\boldsymbol{\beta}}_t)}{1 + \exp(\hat{\alpha}_t + \mathbf{x}_d' \hat{\boldsymbol{\beta}}_t)} \tag{3}$$

An article is classified as *Republican-leaning* if $\hat{P}(\mathrm{R} \mid d) > 0.5$ and *Democratic-leaning* otherwise.

These scores measure how closely an article's vocabulary resembles that of Republican versus Democratic members of Congress. A high $\hat{P}(\mathrm{R} \mid d)$ indicates that an article uses words and phrases favored by Republican legislators, not that it endorses Republican policies.

The top partisan features (Table 12 in Appendix F) reveal clear thematic clusters: Republican-associated language centers on anti-government rhetoric (*liberal*, *bureaucracy*, *big government*), fiscal conservatism (*tax increase*), and social traditionalism (*unborn*, *marriage*); Democratic-associated language emphasizes inequality (*wealthiest*, *tax break*), social programs (*Head Start*), and corporate accountability (*oil company*). These patterns align closely with Gentzkow and Shapiro (2010) and Gentzkow et al. (2019), and make clear why newspaper coverage of a community experiencing trade-induced manufacturing decline would score higher on the Republican dimension.

4.4 Aggregation to Newspaper–Year Panel

I aggregate article-level classifications to a newspaper–year panel. The primary outcome is *Share R-leaning*: the fraction of articles with $\hat{P}(\mathrm{R} \mid d) > 0.5$. Because every article is classified as either R- or D-leaning at the 0.5 threshold, Share D-leaning = 1 − Share R-leaning by construction. I use this binary share rather than the continuous mean score because the binary indicator is robust to extreme values in the tails of the article-level score distribution, which can introduce noise in the continuous mean without reflecting meaningful changes in partisan language.

External validation confirms that the measure recovers meaningful cross-sectional vari-

ation: the correlation between a newspaper's average share of R-leaning articles and the pre-NAFTA Republican vote share in its home county is $r = 0.35$ ($p < 0.001$; Figure 7 in Appendix D), consistent with the demand-side validation in Gentzkow and Shapiro (2010). The measure is also stable over time within newspapers (year-to-year autocorrelation $r = 0.80$; Figure 6 in Appendix D), and aggregation over thousands of articles per newspaper–year yields highly precise estimates (Appendix D).

A natural concern is whether partisan language identified in congressional speeches transfers meaningfully to newspaper text. Three features of the design address this: the shared vocabulary filter restricts the feature space to terms that appear in both corpora; the vote share correlation confirms that the projected measure recovers real cross-sectional variation; and, most importantly, the identification strategy relies entirely on *within-newspaper, over-time* variation (the 25% residual in the variance decomposition), so the measure need only track changes in a given newspaper's partisan language, not rank newspapers accurately in cross-section. The balance test (Table 1) confirms that trade vulnerability is orthogonal to pre-NAFTA media language.

4.5 Summary Statistics

Table 2 reports summary statistics for the estimation sample.

Table 2: Summary Statistics

	N	Mean	SD	P25	P50	P75
Panel A: Outcome						
Share R-leaning	2077	0.532	0.090	0.468	0.554	0.593
Panel B: Treatment and controls						
NAFTA vulnerability (scaled)	2077	0.293	0.159	0.197	0.240	0.364
China import shock	2077	1.026	0.658	0.551	0.917	1.272
Manuf. empl. share, 1990	2077	0.211	0.073	0.164	0.208	0.257
Panel C: Commuting zone characteristics						
Share bachelor's or higher, 1990	2077	0.206	0.049	0.173	0.205	0.232
Per capita income, 1989 ($1000s)	2077	43.3	7.2	37.9	42.8	47.1
Articles per newspaper–year (1000s)	2077	25.3	17.2	12.7	21.3	33.6

Notes: The estimation sample covers 131 daily newspapers in 98 commuting zones, 1987–2004 (2,077 newspaper–year observations; 52.6 million articles). Share R-leaning denotes the fraction of articles classified as Republican-leaning ($\hat{P}(\mathrm{R} \mid d) > 0.5$); the complement is Share D-leaning. NAFTA vulnerability is the employment-and-RCA-weighted average tariff on Mexican imports, scaled by the Q4–Q1 interquartile difference. China import shock is the Autor et al. (2013) measure of import penetration from China (in $1000s per worker). Per capita income is in 1989 dollars.

The estimation sample is an unbalanced panel of 131 daily newspapers in 98 commuting zones over 1987–2004 (2,077 newspaper–year observations; 52.6 million articles). All 131 newspapers appear by 1992; earlier years contain fewer papers (49 in 1987) as some entered the digital archive later (Table 8 in Appendix E). A variance decomposition of Share R attributes 65% of variance to year effects, 10% to newspaper fixed effects, and 25% to the residual—the within-newspaper, within-year variation that identifies the event study (Table 7 in Appendix D).

5 Empirical Strategy

To estimate the causal effect of trade shocks on media slant, I employ a difference-in-differences event study design. I compare the evolution of partisan shares in newspapers located in trade-exposed areas to those in less exposed areas, before and after the implemen-

tation of NAFTA. The baseline specification is:

$$Y_{nc,t} = \sum_{k \neq 1993} \beta_k \left(Vul_c \times \mathbf{1}\{t = k\} \right) + \alpha_n + \gamma_{dt} + \epsilon_{nc,t} \tag{4}$$

where $Y_{nc,t}$ is Share R-leaning for newspaper n in Commuting Zone c in year t. Vul_c is the time-invariant 1990 trade vulnerability of the Commuting Zone, scaled by the gap between top- and bottom-quartile mean vulnerability, so that a one-unit increase corresponds to moving from the average bottom-quartile to the average top-quartile area. α_n denotes newspaper fixed effects, which absorb time-invariant newspaper characteristics. γ_{dt} denotes Census division d by year t fixed effects, which absorb region-specific trends in political language—for example, differential responses to national political events across regions. Identification therefore comes from within-division, within-year variation in trade exposure. Standard errors are clustered at the Commuting Zone level. Vul_c enters the regression directly as a continuous treatment intensity measure, not as an instrumental variable; the design requires no first stage or exclusion restriction.

The coefficients β_k trace the dynamic response of Share R to the trade shock relative to the base year 1993. A finding of $\beta_k \neq 0$ for $k \geq 1994$ would indicate that newspapers in trade-exposed regions adjusted their partisan rhetoric in response to economic integration. The leads ($k < 1993$) serve as a test for parallel trends.

5.1 Threats to Identification

The identifying assumption is that, conditional on newspaper, year, and division-by-year fixed effects, trade vulnerability is uncorrelated with unobserved determinants of media slant trends. Several threats to identification deserve attention.

Shift-share structure and the exogeneity of exposure. The vulnerability measure has a Bartik shift-share structure: it combines pre-determined 1990 industry employment

shares with NAFTA-induced tariff changes (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Identification can rely on either the exogeneity of the shifts (tariff changes) or the exogeneity of the shares (industrial composition). I rely primarily on the former: NAFTA tariff schedules were negotiated at the national level across hundreds of product lines, making it implausible that tariff changes in any particular industry were set in response to the media language of specific commuting zones. The pre-determined industry shares, by contrast, reflect historical patterns of economic specialization that may correlate with other local characteristics, making the exogeneity of shares a stronger assumption. The balance test (Table 1) supports this assumption: pre-treatment media outcomes are uncorrelated with vulnerability.

Differential trends in manufacturing areas. Division-by-year fixed effects absorb nationwide and region-wide shifts in political language. A remaining concern is that manufacturing-heavy areas may have followed different political trends for reasons unrelated to trade. The manufacturing share control addresses this by allowing manufacturing-dependent areas to follow a separate time path, and the China shock control absorbs differential trends from a concurrent trade shock.

Pre-trend power and the unbalanced panel. The estimation sample is unbalanced, with 49 newspapers in 1987 rising to 131 by 1992. Because the pre-NAFTA coefficients for 1987–1991 are estimated from smaller samples, the pre-trend test has lower power in these early years. Two facts mitigate this concern. First, the balance test confirms that vulnerability is orthogonal to pre-NAFTA media outcomes. Second, a joint Wald test of the null hypothesis that all six pre-NAFTA coefficients ($\beta_{1987}, \ldots, \beta_{1992}$) equal zero fails to reject ($p = 0.79$).[2]

[2]The test has $k = 6$ restrictions and uses the CRV1 cluster-robust variance–covariance matrix with 98 commuting zone clusters.

6 Results

6.1 Share of Partisan Articles

Figure 2 presents the event study for Share R-leaning. The pre-NAFTA coefficients are close to zero and statistically insignificant, consistent with parallel trends. After 1994, Share R rises sharply: trade-exposed newspapers produce a progressively larger share of Republican-leaning articles. The effect reaches approximately 0.10 by 1997 ($p < 0.01$) in the controlled specification and remains elevated through 2004. The results are robust to controlling for the China import shock and manufacturing employment share (blue squares), which leave the overall pattern intact.

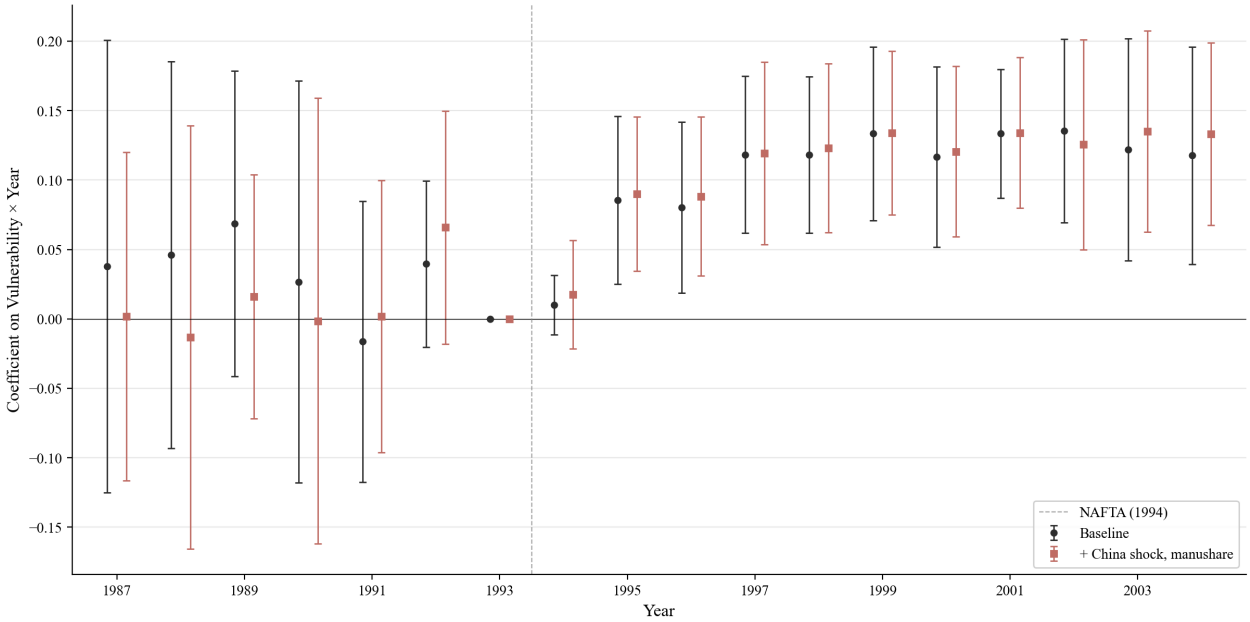

Figure 2: Event Study: Share of R-Leaning Articles

Notes: Each point plots the coefficient on $Vul_c \times \mathbf{1}\{t = k\}$ with Share R-leaning as the dependent variable. Black circles show the baseline specification; blue squares add China shock and manufacturing share controls interacted with year. Vertical bars indicate 95% confidence intervals based on standard errors clustered at the commuting zone level. The dashed line marks 1993, the omitted base year.

6.2 Pooled Difference-in-Differences

Table 3 reports the pooled difference-in-differences estimates. Columns (1)–(3) collapse the pre- and post-NAFTA periods into a single treatment indicator $Post_t = \mathbf{1}\{t \geq 1994\}$. Column (1) includes newspaper, year, and division-by-year fixed effects. Column (2) adds 1990 manufacturing employment share interacted with year dummies to absorb secular trends in manufacturing-dependent areas. Column (3) further controls for the China import shock interacted with year. Columns (4)–(5) decompose the post-NAFTA period into a short-term effect (1994–1998) and a long-term effect (1999–2004), using the full control set of column (3).

Share R-leaning increases by 9.6 percentage points for a top-versus-bottom-quartile shift in vulnerability ($p < 0.01$). Importantly, the shift reflects a change in the partisan *composition* of articles, not an increase in the overall level of partisan language. Replacing the outcome with *politicization*—the mean absolute deviation of article-level scores from the partisan midpoint, which captures the total intensity of partisan language regardless of direction—yields an insignificant coefficient (0.095, $p = 0.28$). Trade-exposed newspapers did not produce more partisan language overall; the *balance* of that language shifted toward Republican-leaning terms.

The estimate is unchanged when the China import shock is added as a control (columns 2 and 3), indicating that the NAFTA effect is not confounded by the concurrent rise in Chinese imports.

Columns (4)–(5) decompose the post-NAFTA period. In the short term (1994–1998), Share R increases by 7.2 percentage points ($p < 0.05$); in the long term (1999–2004), the effect reaches 11.5 percentage points ($p < 0.01$). This pattern is consistent with the event study dynamics in Figure 2, where the coefficients rise through the late 1990s and remain elevated through 2004.

Table 3: Difference-in-Differences: NAFTA Vulnerability and Media Slant

	(1)	(2)	(3)	(4)	(5)
		Pooled		*Short-term*	*Long-term*
Share R-leaning	0.084***	0.096***	0.096***	0.072**	0.115***
	(0.028)	(0.032)	(0.032)	(0.029)	(0.035)
Paper FE	Yes	Yes	Yes	Yes	Yes
Division × Year FE	Yes	Yes	Yes	Yes	Yes
Manuf. share × Year		Yes	Yes	Yes	Yes
China shock × Year			Yes	Yes	Yes
Observations	2,077	2,077	2,077	2,077	

Notes: Columns (1)–(3) report the coefficient on $Vul_c \times Post_t$ from separate regressions, where $Post_t = \mathbf{1}\{t \geq 1994\}$. Columns (4)–(5) come from a single regression that replaces $Post_t$ with two indicators: $Short_t = \mathbf{1}\{1994 \leq t \leq 1998\}$ and $Long_t = \mathbf{1}\{t \geq 1999\}$, using the full control set of column (3). The sample covers 1987–2004. "Share R-leaning" is the fraction of articles with $\hat{P}(\text{R} \mid d) > 0.5$; the complement (Share D-leaning = 1 − Share R-leaning) is omitted. NAFTA vulnerability is scaled by the Q4–Q1 interquartile difference, so coefficients represent the effect of moving from a low- to a high-exposure commuting zone. Standard errors clustered at the commuting zone level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The direction of the rightward shift has a natural explanation. Trade policy in the 1990s was cross-cutting—a Republican president initiated NAFTA and a Democratic one signed it—but the *language* associated with trade-affected economic concerns maps asymmetrically onto partisan rhetoric. After the 1994 Republican Revolution, partisan congressional vocabulary diverged sharply (Gentzkow et al., 2019), with Republican rhetoric increasingly associated with themes—such as skepticism of government programs and fiscal conservatism—that overlapped with the economic anxieties of trade-affected communities. The rightward shift therefore does not reflect a change in which party's policy position newspapers adopt. Rather, trade shocks increased the salience of topics whose associated language was disproportionately Republican-associated in Congress.

6.3 The Shift Is Broad, Not Trade-Specific

A natural question is whether the rightward shift is concentrated in trade-related content or distributed broadly across topics. Three pieces of evidence point to the latter.

First, trade-specific features carry negligible weight in the partisan vocabulary. I classify all LASSO features into eleven substantive categories and compute each category's share of the total absolute coefficient weight across all nine Congress windows. Trade-related language (e.g., *trade*, *tariff*, *nafta*, *export*, *quota*, *wto*) accounts for just 0.3% of the total absolute coefficient weight—the smallest category. The largest substantive categories are Government & Procedure (5.4%), Tax & Fiscal (4.4%), and Defense & Foreign (1.7%); the remaining 83% of the coefficient weight falls on general-purpose partisan terms not tied to any specific policy domain. None of the 50 most influential features (ranked by average absolute coefficient) is trade-related (Table 17 in Appendix G). This reflects the fact that trade was a cross-cutting issue during the NAFTA era—both parties had vocal supporters and opponents—so trade-specific language does not consistently distinguish Republican from Democratic rhetoric.

Second, I directly test whether trade-related language drives the results by zeroing all trade-related coefficients in the LASSO models and re-scoring all newspaper articles from scratch. Under a narrow definition (26 features: *trade*, *tariff*, *nafta*, *export*, *gatt*, *wto*, *quota*, and their bigrams), these features account for 0.4% of the total coefficient weight. Zeroing them and re-estimating the full DiD changes the Share R coefficient by less than 0.5% (from 0.096 to 0.096). Under a broad definition (81 features, including all forms of *import*), the weight rises to 0.9% and the estimate changes by less than 0.5% (Table 18 in Appendix G). The compositional shift is entirely attributable to general-purpose partisan language—not trade content.

Third, I directly verify that the compositional shift is broad by computing the share of R-leaning articles *within* each topic category and re-estimating the DiD separately for each. I classify articles as mentioning a topic if any of the topic's keyword features appear in the article's feature vector; articles can belong to multiple topics. Figure 3 presents the results: all topic categories except Tax & Fiscal show positive coefficients, and the majority are statistically significant. Trade-mentioning articles (3% of the sample) shift by 0.075—

comparable to the overall estimate of 0.096. The largest effects appear in Education (0.116, $p < 0.01$) and articles with no specific topical keywords (0.102, $p < 0.01$, 30% of articles). Tax & Fiscal shows no shift (-0.003, $p = 0.94$), consistent with tax articles already having a well-separated partisan vocabulary that leaves little room for threshold crossing. The fact that the rightward shift appears in nearly every topic category rules out the possibility that trade-specific coverage mechanically drives the results. Moreover, using each topic's *article share* as the outcome in the same DiD specification yields no significant coefficient for any of the ten categories ($p > 0.14$ for all), confirming that the shift operates within topics rather than through a reallocation of coverage across them.

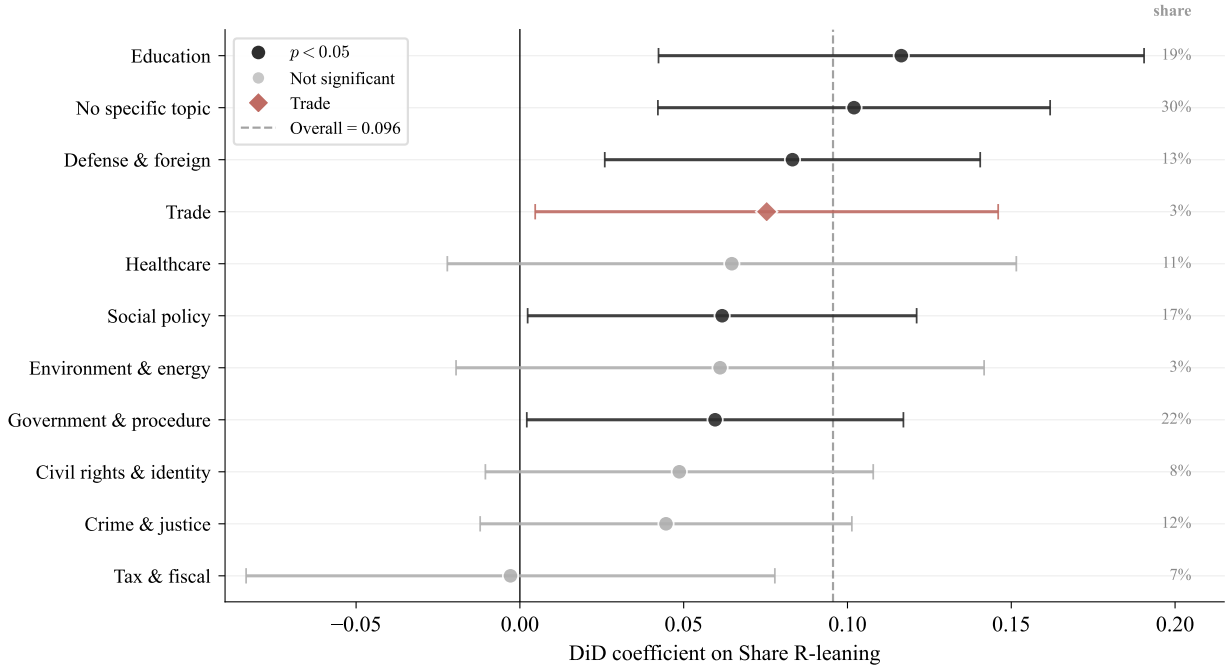

Figure 3: Share of R-Leaning Articles by Topic: DiD Coefficients

Notes: Each bar reports the DiD coefficient on the topic-specific share of R-leaning articles. Error bars show 95% confidence intervals (CZ-clustered). The dashed red line marks the overall coefficient (0.096). Trade articles (highlighted in red) shift comparably to other topics. Tax & Fiscal articles show no shift, consistent with an already well-separated partisan vocabulary. Table 19 in Appendix G reports the full regression results.

6.4 Robustness

The sequential inclusion of controls in Table 3 confirms that confounders do not drive the results. Adding manufacturing share-by-year interactions (column 2) slightly strengthens most coefficients, suggesting that the baseline specification, if anything, understates the effect by not absorbing secular trends in manufacturing-dependent areas. The China import shock control (column 3) barely alters the estimates, consistent with the China shock operating through different industries and timing than NAFTA. The stability of the coefficients across specifications supports a causal interpretation.

6.4.1 Randomization Inference

With 98 commuting zone clusters, asymptotic cluster-robust standard errors may be imprecise. To provide inference that does not rely on asymptotic approximations, I conduct randomization inference by permuting the CZ-level vulnerability assignment 5,000 times. For each permutation, I randomly reassign vulnerability values across CZs (preserving the panel structure within each CZ), reconstruct the treatment variable, and re-estimate the DiD coefficient using the Frisch–Waugh–Lovell theorem for computational efficiency.[3]

The permutation p-value strongly confirms the result: Share R ($p < 0.001$) is highly significant, with the actual coefficient falling well into the tail of the permutation distribution (Figure 13 in Appendix G).

6.4.2 Alternative Training Sample

As a further robustness check, I re-estimate the slant model restricting the training sample to the most partisan 20% of legislators in each Congress (identified by Nokken–Poole scores; see Appendix H for details). By training only on speakers whose partisan identity is most clearly revealed, this specification learns a sharper partisan vocabulary—but from a substan-

[3] I verify that the FWL coefficients match the pyfixest point estimates to machine precision ($< 10^{-14}$) before running the permutation loop.

tially smaller training sample, which may introduce additional estimation noise. The result is virtually identical: Share R increases by 9.8 percentage points ($p < 0.01$; Table 21 in Appendix H). This is a meaningful test: training on the partisan core selects a different—and more extreme—set of discriminating phrases, so the resulting measure captures resemblance to the language of the *most* partisan legislators rather than the average party member. The near-identical estimate indicates that the rightward shift in trade-exposed newspapers extends to this stricter definition of partisan language.

6.4.3 Panel Balance

Because the panel is unbalanced—49 newspapers in 1987 rising to 131 by 1992—I verify that newspaper entry and exit do not drive the results. Restricting the sample to the 46 newspapers present in all 18 years (39 commuting zones, 828 observations), the estimate is virtually unchanged: Share R increases by 11.5 percentage points ($p < 0.01$). Restricting the sample period to 1992–2004, when all 131 newspapers are present (1,663 observations), yields 7.7 percentage points ($p < 0.05$). The main results are not an artifact of the unbalanced panel structure (Table 16 in Appendix G). Moreover, newspaper survival is unrelated to trade exposure: the mean vulnerability of newspapers that span the full sample is statistically indistinguishable from those with partial coverage ($p = 0.84$), and the correlation between vulnerability and entry year is effectively zero ($r = 0.03$, $p = 0.73$).

6.4.4 Classification Threshold

I verify that the share results are not driven by articles near the classification boundary. Applying progressively stricter probability thresholds ($\tau = 0.5, 0.6, 0.7, 0.8, 0.9$), the post-NAFTA rightward shift in Share R survives at all cutoffs, and the R Ratio—the share of R-leaning among all confidently classified articles—*strengthens* at stricter thresholds (from 0.095 at $\tau = 0.5$ to 0.188 at $\tau = 0.9$), confirming that the compositional shift is concentrated among clearly partisan articles rather than centrist noise near the decision boundary

(Figure 10 in Appendix G).

6.5 Heterogeneity

6.5.1 Local Education

If trade shocks reshape the political language of local media, the effect should be stronger in communities where the economic dislocation is most acute. Areas with lower human capital face greater adjustment frictions: workers displaced by trade liberalization have fewer outside options, prolonging economic hardship (Autor et al., 2013; Hakobyan and McLaren, 2016). I test this prediction by interacting the baseline DiD with the 1990 share of adults holding a bachelor's degree or higher in each commuting zone.

Table 4 reports the results. In the median-split specification (column 1), the base coefficient shows that Share R increases by 10.1 percentage points in low-education commuting zones ($p < 0.01$), while the interaction reveals that the effect is 12.6 percentage points weaker in high-education areas ($p < 0.01$)—leaving an implied effect of -0.025 for the high-education group. The continuous specification (column 2) confirms this pattern: a one-standard-deviation increase in local education attenuates the effect by 6.8 percentage points ($p < 0.01$). These patterns suggest that trade shocks generate the strongest media response where economic adjustment is most difficult. In communities with fewer college-educated workers, NAFTA-induced job losses concentrated in manufacturing were harder to absorb and sustained local economic distress for longer.[4][5]

[4]Because education and vulnerability are negatively correlated ($r = -0.37$), the education interaction could in principle capture nonlinearity in the vulnerability effect rather than an independent moderating role of education. Adding $Vul_c^2 \times Post_t$ to the specification rules this out: the quadratic term is insignificant ($p = 0.87$) and the education interaction is unchanged (-0.119, $p < 0.001$).

[5]The effect is also slightly weaker when interacted with newspaper size (mean pre-NAFTA article count), with a continuous interaction of -0.034 ($p < 0.01$) for Share R, suggesting that larger newspapers are marginally less responsive.

Table 4: Heterogeneous Effects by Local Education Level

	(1) *Median split*	(2) *Continuous*
$Vul_c \times Post_t$	0.101***	0.096***
	(0.033)	(0.032)
$Vul_c \times Post_t \times HighEdu_c$	−0.126***	
	(0.030)	
$Vul_c \times Post_t \times Edu_c$		−0.068***
		(0.015)
Implied effect for high-education CZs	−0.025	
Paper FE	Yes	Yes
Division × Year FE	Yes	Yes
China shock × Year	Yes	Yes
Manuf. share × Year	Yes	Yes
Observations	2,077	2,077

Notes: Column (1) augments the baseline DiD with $Vul_c \times Post_t \times HighEdu_c$, where $HighEdu_c$ equals one for commuting zones with above-median share of adults holding a bachelor's degree or higher in 1990 (median = 20.4%; 66 low-education and 65 high-education papers). The base coefficient captures the NAFTA effect in low-education areas; the interaction captures the differential for high-education areas. The implied effect in high-education areas is the sum ($0.101 - 0.126 = -0.025$). Column (2) interacts with Edu_c, the standardized share of adults with a bachelor's degree or higher (SD = 5.0%); the coefficient represents the change in the NAFTA effect per one-standard-deviation increase in local education. All specifications include the full set of fixed effects and controls from column (3) of Table 3. Standard errors clustered at the commuting zone level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5.2 Pre-NAFTA Slant: Convergence, Not Amplification

The education results show that trade shocks produce stronger media responses in economically vulnerable communities. A complementary question is whether the rightward shift amplifies pre-existing partisan leanings or instead produces convergence across newspapers with different initial positions. If NAFTA exposure amplifies existing tendencies, initially Republican-leaning papers should shift further rightward. If instead the shock generates convergence, initially Democratic-leaning papers should exhibit the largest response.

Table 5 tests this by interacting the baseline DiD with each newspaper's pre-NAFTA mean share of R-leaning articles (average over 1987–1993). The results strongly favor con-

vergence. In the median-split specification (column 1), the base coefficient shows that Share R increases by 11.1 percentage points for initially D-leaning papers ($p < 0.01$). The interaction is large and negative (-0.146, $p < 0.01$), so the implied effect for initially R-leaning papers is effectively zero ($0.111 - 0.146 = -0.035$). The continuous specification (column 2) reinforces this: a one-standard-deviation increase in pre-NAFTA slant reduces the effect by 13.6 percentage points ($p < 0.01$)—nearly wiping out the baseline effect. The overall picture is one of *compositional convergence*, not amplification.[6]

A natural concern is that the convergence pattern partly reflects mechanical regression to the mean: newspapers classified as D-leaning in the pre-period have low Share R by construction, so any positive shock would produce larger measured increases for them. Two features of the analysis mitigate this concern. First, if regression to the mean were driving the result, it should affect D-leaning papers *everywhere*—a paper with pre-NAFTA Share R of 0.40 in New York should revert just as much as one with the same initial value in Detroit, since measurement noise does not depend on geography. But the convergence finding is an *interaction* with trade vulnerability: among papers with identical initial slant, those in high-exposure commuting zones shift substantially while those in low-exposure areas barely move. This geographic dependence is inconsistent with a purely mechanical story. Second, the division×year fixed effects already absorb any common or region-specific rightward drift, so the identifying variation is within-division-year—leaving no common shock to drive mechanical reversion.

The convergence pattern also speaks to the mechanism. If the effect operated primarily through the supply side—for example, if the loss of local advertising clients forced newspapers to alter content—one would expect similar responses regardless of initial partisan lean, since advertising revenue declines similarly for initially D-leaning and R-leaning papers.[7] Instead,

[6]Because initially D-leaning papers tend to be located in slightly lower-education, higher-vulnerability areas ($r = 0.19$ between pre-NAFTA slant and education), the convergence pattern could in principle reflect the education heterogeneity documented above. Including both interactions simultaneously rules this out: the convergence interaction remains strong (-0.131, $p < 0.001$) and the education interaction remains significant (-0.089, $p = 0.001$), confirming that the two effects are independently identified.

[7]Splitting the advertising long-difference regressions by pre-NAFTA median partisan lean yields similar

the shift concentrates entirely in D-leaning papers—consistent with a demand-side channel. Choi et al. (2024) show that historically Democratic, trade-exposed counties shifted toward the Republican Party after NAFTA, and newspapers in these areas would face the strongest pressure to adjust their content to match changing reader preferences.

Table 5: Heterogeneous Effects by Pre-NAFTA Slant

	(1) *Median split*	(2) *Continuous*
$Vul_c \times Post_t$	0.111^{***}	0.096^{***}
	(0.031)	(0.032)
$Vul_c \times Post_t \times HighSlant_i$	-0.146^{***}	
	(0.030)	
$Vul_c \times Post_t \times PreSlant_i$		-0.136^{***}
		(0.014)
Implied effect for R-leaning papers	-0.035	
Paper FE	Yes	Yes
Division × Year FE	Yes	Yes
China shock × Year	Yes	Yes
Manuf. share × Year	Yes	Yes
Observations	2,077	2,077

Notes: Column (1) augments the baseline DiD with $Vul_c \times Post_t \times HighSlant_i$, where $HighSlant_i$ equals one for newspapers with above-median pre-NAFTA share of R-leaning articles (mean over 1987–1993; 66 initially D-leaning and 65 initially R-leaning papers). The base coefficient captures the NAFTA effect for initially D-leaning papers; the interaction captures the differential for initially R-leaning papers. The implied effect for R-leaning papers is the sum ($0.111 - 0.146 = -0.035$). Column (2) interacts with $PreSlant_i$, the standardized pre-NAFTA mean share of R-leaning articles (SD = 0.05); the coefficient represents the change in the NAFTA effect per one-standard-deviation increase in initial slant. All specifications include the full set of fixed effects and controls from column (3) of Table 3. Standard errors clustered at the commuting zone level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

This paper shows that trade shocks reshape the political language of local news. Newspapers in commuting zones more exposed to NAFTA underwent a systematic rightward shift: the

point estimates for both groups (e.g., −$56 for D-leaning vs. −$48 for R-leaning at the 1988–1996 horizon), confirming that the advertising channel does not differentially affect one group.

share of Republican-leaning articles increased by 9.6 percentage points for a top-versus-bottom-quartile difference in trade vulnerability ($p < 0.01$, permutation $p < 0.001$). The effect emerges after NAFTA and persists through the sample period, concentrates in low-education communities, and is not driven by trade-specific vocabulary—the shift instead operates through general-purpose partisan language distributed across virtually all article categories.

The convergence pattern provides the clearest evidence on mechanism. Initially Democratic-leaning newspapers drive the entire rightward shift, while initially Republican-leaning papers are unaffected. A pure supply-side story—in which the loss of local advertising clients forces content changes—cannot explain why the response depends on a newspaper's initial partisan lean. The pattern is instead consistent with a demand-side channel. Choi et al. (2024) document that trade-exposed communities turned away from the Democratic Party that signed NAFTA; the convergence finding suggests that newspapers in these areas adjusted their content to match changing reader preferences. The fact that convergence scales with trade vulnerability—rather than appearing uniformly among all D-leaning papers—rules out mechanical regression to the mean and reinforces the link to trade exposure.

These findings contribute to the literature in two ways. First, while existing work identifies reader ideology, ownership, and competition as determinants of media slant (Gentzkow and Shapiro, 2010; Mullainathan and Shleifer, 2005; Gentzkow et al., 2011), this paper documents that an exogenous economic shock also moves media language—and that the demand-side response of readers is the most likely channel. Second, the results suggest that local media may amplify the political consequences of trade liberalization documented by Autor et al. (2020) and Choi et al. (2024). Residents of trade-exposed communities face a systematically different information environment: not because their newspapers cover trade differently, but because the share of Republican-leaning articles increases broadly across topics. Given the evidence that media language shapes political attitudes (DellaVigna and Kaplan, 2007; Martin and Yurukoglu, 2017), this shift is a plausible contributor to the

durable political effects of trade liberalization. While NAFTA provides a uniquely clean identification setting—local newspapers held a near-monopoly on political information before the internet—the core finding is that real economic shocks reshape the political language available to local audiences. This mechanism does not depend on the newspaper format and could operate through any medium that serves as a local information channel.

Three limitations deserve note. First, while the convergence pattern favors a demand-side interpretation, the analysis cannot definitively rule out all supply-side contributions. The convergence test shows that D-leaning papers drive the shift and that this depends on trade exposure—which is inconsistent with a pure supply-side story—but a fully causal mechanism test would require exogenous variation in reader preferences separate from trade exposure, which is not available. Second, the paper documents a reduced-form link between trade exposure and media language but cannot establish that the media shift causally affected downstream political outcomes. Doing so would require an instrument that shifts media language without directly affecting political attitudes—and trade vulnerability, which drives both, does not satisfy this exclusion restriction. Third, the sample of 131 newspapers in 98 commuting zones limits statistical power for subgroup analyses and yields an unbalanced panel—with fewer newspapers in the early pre-NAFTA years—that weakens pre-trend tests.

References

Autor, D., D. Dorn, G. Hanson, and K. Majlesi (2020). Importing political polarization? the electoral consequences of rising trade exposure. *American Economic Review 110*(10), 3139–3183.

Autor, D. H., D. Dorn, and G. H. Hanson (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review 103*(6), 2121–2168.

Borusyak, K., P. Hull, and X. Jaravel (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies 89*(1), 181–213.

Choi, J., I. Kuziemko, E. Washington, and G. Wright (2024). Local economic and political effects of trade deals: Evidence from nafta. *American Economic Review 114*(6), 1540–1575.

DellaVigna, S. and E. Kaplan (2007). The fox news effect: Media bias and voting. *The Quarterly Journal of Economics 122*(3), 1187–1234.

Dix-Carneiro, R., R. R. Soares, and G. Ulyssea (2018). Economic shocks and crime: Evidence from the brazilian trade liberalization. *American Economic Journal: Applied Economics 10*(4), 158–195.

Dorn, D. (2009). Essays on inequality, spatial interaction, and the demand for skills. Dissertation, University of St. Gallen.

Feenstra, R. C., J. Romalis, and P. K. Schott (2002). Us imports, exports, and tariff data, 1989-2001.

Feler, L. and M. Z. Senses (2017). Trade shocks and the provision of local public goods. *American Economic Journal: Economic Policy 9*(4), 101–143.

Gentzkow, M. and J. M. Shapiro (2006). Media bias and reputation. *Journal of Political Economy 114*(2), 280–316.

Gentzkow, M. and J. M. Shapiro (2010). What drives media slant? evidence from us daily newspapers. *Econometrica 78*(1), 35–71.

Gentzkow, M., J. M. Shapiro, and M. Sinkinson (2011). The effect of newspaper entry and exit on electoral politics. *American Economic Review 101*(7), 2980–3018.

Gentzkow, M., J. M. Shapiro, and M. Taddy (2019). Measuring group differences in high-dimensional choices: Method and application to congressional speech. *Econometrica 87*(4), 1307–1340.

Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020). Bartik instruments: What, when, why, and how. *American Economic Review 110*(8), 2586–2624.

Hakobyan, S. and J. McLaren (2016). Looking for local labor market effects of nafta. *Review of Economics and Statistics 98*(4), 728–741.

Laver, M., K. Benoit, and J. Garry (2003). Extracting policy positions from political texts using words as data. *American Political Science Review 97*(2), 311–331.

Lewis, J. B., K. Poole, H. Rosenthal, A. Boche, A. Rudkin, and L. Sonnet (2024). Voteview: Congressional roll-call votes database. https://voteview.com/.

Lu, Y., X. Shao, and Z. Tao (2018). Exposure to chinese imports and media slant: Evidence from 147 U.S. local newspapers over 1998–2012. *Journal of International Economics 114*, 316–330.

Martin, G. J. and A. Yurukoglu (2017). Bias in cable news: Persuasion and polarization. *American Economic Review 107*(9), 2565–2599.

Monroe, B. L., M. P. Colaresi, and K. M. Quinn (2008). Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis 16*(4), 372–403.

Mullainathan, S. and A. Shleifer (2005). The market for news. *American Economic Review 95*(4), 1031–1053.

Petrova, M. (2012). Newspapers and parties: How advertising revenues created an independent press. *American Political Science Review 106*(4), 790–808.

Pierce, J. R. and P. K. Schott (2020). Trade liberalization and mortality: Evidence from US counties. *American Economic Review: Insights 2*(1), 47–64.

Taddy, M. (2013). Multinomial inverse regression for text analysis. *Journal of the American Statistical Association 108*(503), 755–770.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological) 58*(1), 267–288.

Tolbert, C. M. and M. Sizer (1996). US commuting zones and labor market areas: A 1990 update. Technical Report Staff Paper 9614, Economic Research Service, U.S. Department of Agriculture.

Trefler, D. (2004). The long and short of the Canada–US free trade agreement. *American Economic Review 94*(4), 870–895.

Widmer, P., S. Galletta, and E. Ash (2025). Media slant is contagious. *The Economic Journal*.

A Data Sources and Sample Construction

A.1 Newspaper Data

My primary data source for local news content is NewsLibrary, which contains full-text archives of thousands of U.S. newspapers. To construct the analysis sample, I first standardize newspaper names and link them to their headquarters' county FIPS codes by extracting city and state names from publication titles using regex patterns, supplemented by manual headquarters searches for papers whose titles lack geographic identifiers. I restrict the sample to English-language newspapers located in the continental United States.

To identify relevant political news articles, I apply a title-based keyword filter that removes non-news content: death notices and obituaries, birth and wedding announcements, classifieds and legal notices, lifestyle sections (recipes, horoscopes, comics, puzzles, TV/movie listings), photo captions, weather reports, and church listings. An article is excluded if its title matches any of these case-insensitive patterns, removing 2,033,746 articles (3.6% of the post-standardization sample).

I retain newspapers that publish at least 100 articles per year in each of 1992–1995—ensuring consistent coverage around the NAFTA implementation date—and appear for at least six consecutive years within the sample period (1987–2004). I exclude national newspapers (USA Today and Christian Science Monitor) whose content reflects national rather than local coverage, making them unsuitable for a local trade-shock identification strategy. I also exclude four non-daily publications (one alternative weekly, one Sunday-only edition, one semi-weekly, and one paper with severely incomplete digital archives) to maintain a homogeneous sample of daily newspapers. The resulting panel is unbalanced: all 131 newspapers are present by 1992, but earlier years contain fewer papers as some entered the digital archive after 1987, and a small number exit toward the end of the sample. Merged or renamed newspapers are treated as distinct outlets. The final estimation sample comprises 131 daily newspapers in 98 commuting zones (Table 9 in Appendix E).

Table 6 documents the sample attrition from the initial NewsLibrary universe to the final estimation panel.

Table 6: Sample Construction

Step	Articles	Newspapers	CZs
Initial NewsLibrary universe (1987–2004)	105,818,269		
After name standardization & target selection	55,894,076	138	
After non-news content filtering	53,860,330	138	
After excluding national newspapers	52,855,686	136	
After excluding non-daily publications	52,772,399	132	
After FIPS linking and CZ assignment	52,772,399	132	99
After requiring non-missing NAFTA vulnerability	52,604,914	131	98

Notes: Each row shows the count remaining after each sample restriction. Target selection retains 138 newspapers that publish at least 100 articles per year in each of 1992–1995, excluding wire services and non-newspaper sources. Non-news filtering removes obituaries, classifieds, television listings, and other non-editorial content based on title keyword patterns (2,033,746 articles removed). Two national newspapers (USA Today and Christian Science Monitor) are excluded because their content reflects national rather than local coverage. Four non-daily publications (one alternative weekly, one Sunday-only edition, one semi-weekly, and one paper with severely incomplete digital archives) are excluded to maintain a homogeneous sample of daily newspapers. FIPS linking assigns each newspaper to a county and commuting zone by extracting city–state pairs from publication titles; one commuting zone (one newspaper) is dropped due to missing 1990 NAFTA vulnerability data. The final estimation sample spans 1987–2004 (2,077 newspaper–year observations). The contemporaneous China import shock is controlled for directly in the regression specifications rather than through sample restriction.

A.2 Congressional Speech Data

Congressional speech data come from the Hein-Bound corpus of floor speeches (Gentzkow et al., 2019), which I restrict to the U.S. House of Representatives for Congresses 99 through 108 (1985–2004). I merge these with legislator ideology data from Voteview (Lewis et al., 2024). I match Hein-Bound speakers to Voteview records using normalized last names, state, and congress, disambiguating ties via first-name token overlap.

A.3 Economic Data

Economic data sources include: (1) tariff data from Feenstra et al. (2002) (1990–2001) and the USITC Tariff Database (2002–2008); (2) import data from Hakobyan and McLaren (2016) and unit-value data from the USITC DataWeb for computing ad valorem equivalent

tariffs; (3) export data from Hakobyan and McLaren (2016) for computing revealed comparative advantage; (4) County Business Patterns (CBP) for local employment by industry (1986–2008); (5) county-level demographic, education, and income controls from Census sources; (6) a county-to-commuting-zone crosswalk (Tolbert and Sizer, 1996; Dorn, 2009); and (7) the Editor & Publisher Yearbook (ICPSR 30261), which provides newspaper-level advertising rates (dollars per column-inch), circulation, and ownership identifiers at five snapshot years (1988, 1992, 1996, 2000, 2004). I match newspapers to the E&P database using county FIPS codes and fuzzy name matching (SequenceMatcher threshold ≥ 0.4), supplemented by manual matches for 14 papers with non-standard names. Of the 131 estimation-sample newspapers, 128 match to E&P (98%). Ownership classification uses 1988 records: a newspaper is *chain-owned* if its owner holds multiple papers in the full E&P universe and *independent* otherwise.

B Construction of Ad Valorem Equivalent Tariffs

I construct the ad valorem equivalent (AVE) tariff for each HS8 product–year by combining the ad valorem rate with the specific rate divided by unit value, drawing on Feenstra et al. (2002) for 1990–2001 and the USITC Tariff Database for 2002–2008. The unit-value data come from the USITC DataWeb and are merged at the HS8–year–quantity-unit level. For products with only specific duties, the AVE equals the specific rate divided by the unit value; for products with both ad valorem and specific components, the AVE is the sum.

I then compute an import-weighted tariff at the SIC industry level, using 1990 Mexican import values as weights. The HS6-to-SIC concordance follows the crosswalk from Dorn (2009). Products with sentinel tariff values (indicating missing or suppressed data in the USITC files) are excluded from the tariff calculation but retained for weight normalization.

C Text Preprocessing Details

The preprocessing pipeline applies the following steps sequentially:

1. **Parliamentary phrase removal.** I remove an expanded list of procedural phrases (e.g., "yield back," "without objection," "motion to table," "amendment thereto," "extension of remarks," "demand a recorded vote") via a single regex pass over the raw speech text.

2. **Tokenization.** Text is tokenized to lowercase alphabetic tokens of at least two characters.

3. **Stop word removal.** Standard English stop words are removed, along with procedural/legal terms ("thereto," "thereof," "hereby," "whereas," etc.) that carry no partisan signal.

4. **Geographic term removal.** I filter out state names, two-letter state abbreviations, and major U.S. city names to ensure the measure captures national political rhetoric rather than local references.

5. **Legislator name removal.** I filter out legislator full-name bigrams (e.g., "newt gingrich"), including nickname variants (e.g., "tip oneill"), constructed from Voteview records for Congresses 99–108. Distinctive legislator surnames that are not common English words (e.g., "dannemeyer" but not "young") are additionally removed as stop words. Individual name tokens that are common English words (e.g., "young," "price," "long") are retained.

6. **Stemming.** All remaining tokens are stemmed using the Porter stemmer.

7. **N-gram construction.** The final feature set consists of stemmed unigrams and bigrams. Unigrams capture individual words with strong partisan connotations (e.g.,

"wealthiest," "bureaucraci," "unborn"), while bigrams preserve multi-word contextual meaning (e.g., "tax relief" vs. "tax break").

8. **Frequency filter.** I apply a per-party frequency filter: a feature is retained if it appears in at least 0.1% of either party's speeches *and* in at least 0.01% of both parties' speeches. This removes extremely rare terms while retaining features that are common in one party's vocabulary even if infrequent in the other's.

9. **Shared vocabulary restriction.** The feature space is restricted to terms that appear in both the congressional speech corpus and the newspaper corpus, with a minimum document frequency of 0.001% in the newspaper corpus. This ensures the LASSO selects only features that are observable in newspaper text, automatically excluding congressional jargon and procedural language that never appears in newspaper articles.

Each individual floor speech serves as a separate document. I apply a count-based vectorizer with L1 row normalization (relative frequencies) and the shared vocabulary restriction described above, which eliminates rare idiosyncratic terms while preserving broadly used partisan vocabulary. The same vectorizer (fitted on congressional text) is applied to newspaper articles to ensure a common feature space.

D Measurement Precision and Validation

The speech-level classification accuracies reported in Table 11 (68–76%) reflect the inherent difficulty of classifying individual floor speeches, many of which address non-partisan topics or contain mixed partisan cues. This appendix demonstrates that (i) aggregation resolves the speech-level noise, (ii) the same convergence pattern carries over from congressional text to newspaper text, (iii) the resulting newspaper-level estimates are temporally stable and externally valid, and (iv) the identification strategy requires only that the measure track *within-newspaper changes* over time—a far weaker requirement than getting absolute partisan levels right.

D.1 Signal Accumulation in Congressional Speech

Figure 4 plots classification accuracy as a function of the number of speeches aggregated per legislator. For each Congress, I draw random subsets of k speeches per legislator, average the predicted probabilities, classify the legislator as Republican if the mean exceeds 0.5, and compute accuracy across all legislators with at least k speeches (200 bootstrap replications per k). At $k = 1$, accuracy matches the speech-level rate (68–76%). By $k = 10$, accuracy exceeds 90% in every Congress. At $k = 50$—well below the median of 85–157 speeches per legislator—accuracy exceeds 97% in most Congresses. The rapid convergence confirms that the LASSO captures a genuine partisan signal: individual documents are noisy, but the signal accumulates reliably with aggregation.

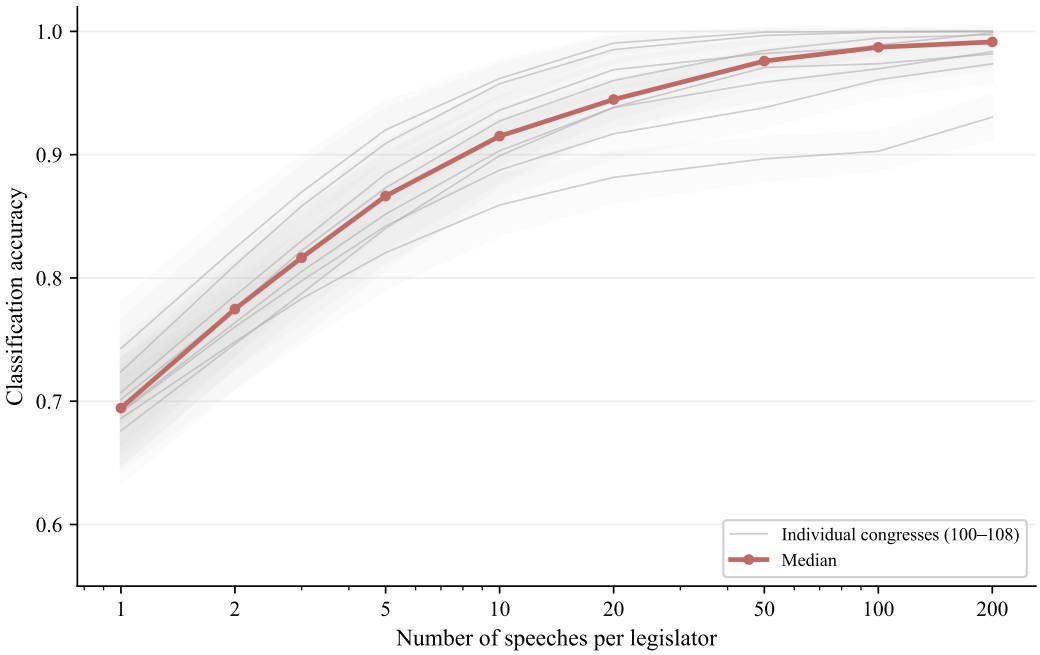

Figure 4: Classification Accuracy by Number of Speeches Aggregated

Notes: Each curve shows the mean classification accuracy when k randomly sampled speeches per legislator are aggregated (200 bootstrap draws). Only legislators with at least k speeches are included. The black line shows the median across Congresses. Shaded bands indicate 95% bootstrap confidence intervals.

D.2 Convergence Across Domains

The same convergence logic applies when the congressional coefficients are projected onto newspaper text. Each newspaper–year aggregates over thousands of article-level classifications, so by the law of large numbers the share of R-leaning articles converges to a stable value. The median newspaper–year in the estimation sample contains approximately 21,300 articles (interquartile range: 12,700–33,600).

Figure 5 demonstrates this directly. For four newspapers spanning the slant distribution, I randomly subsample k articles and compute Share R-leaning, repeating 30 times at each k. At small sample sizes ($k < 200$), individual draws produce dispersed estimates, but by $k \approx 1{,}000$ all traces converge tightly around the full-sample mean. The shaded region marks the interquartile range of actual article counts in the sample, confirming that every newspaper–year observation lies well within the convergence region. The key point is that the convergence pattern established within congressional speech—where party labels provide ground truth—extends seamlessly to the newspaper domain, even though no ground truth exists there.

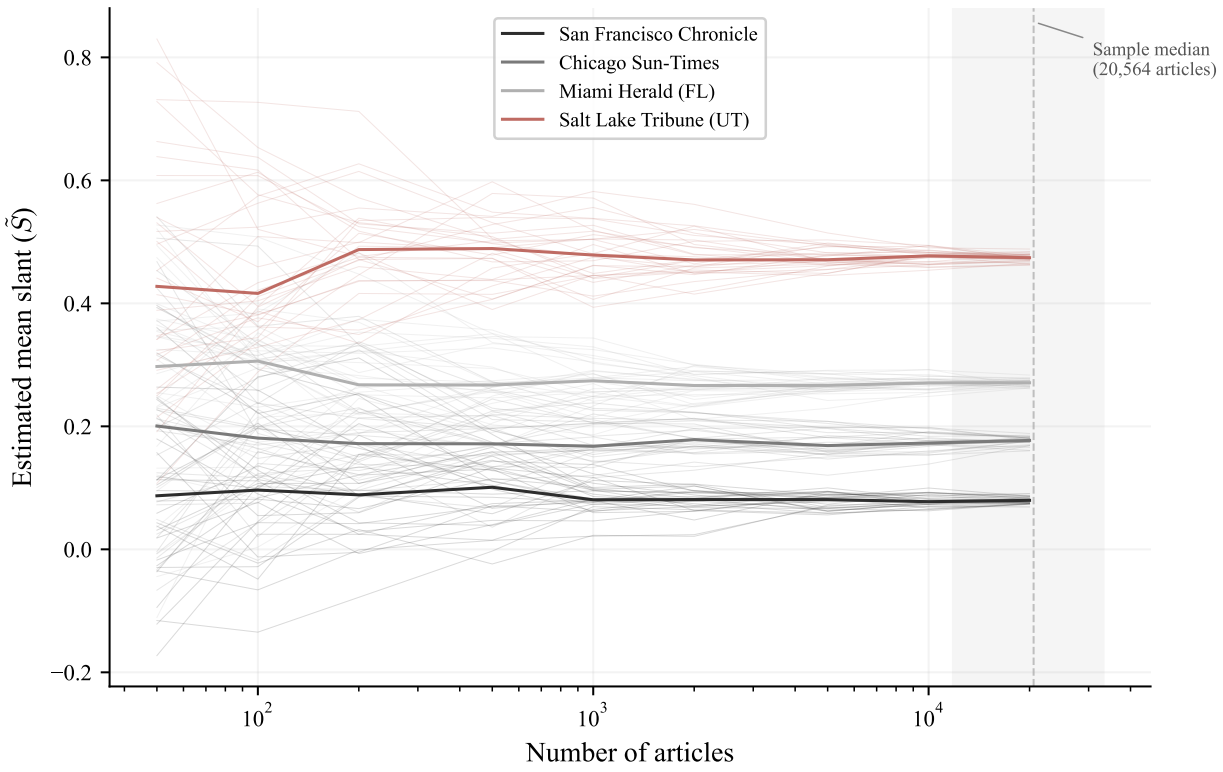

Figure 5: Convergence of Newspaper–Year Mean Slant Estimates

Notes: For each newspaper, 30 random subsamples of size k are drawn from all articles in the newspaper–year (Congress 104). Each thin line traces one subsample's cumulative mean as k increases; the bold line marks the central path. At small k, estimates scatter widely; by $k \approx 1{,}000$, all traces converge to the full-sample mean. The shaded band marks the interquartile range of actual newspaper–year article counts across the full sample (12,700–33,600); the dashed line marks the median (21,300).

D.3 Temporal Stability

If the converged estimates reflect a genuine and persistent feature of each newspaper's language, they should be stable over time. Figure 6 confirms this in two ways. Panel (a) plots each newspaper–year's Share R-leaning against the same newspaper's value in the preceding year: the year-to-year autocorrelation is $r = 0.80$, indicating short-run stability. However, high autocorrelation alone does not rule out slow drift. Panel (b) therefore computes each

newspaper's average over odd and even years separately: the split-half correlation is $r = 0.76$ ($p < 0.001$), confirming that a newspaper's slant is stable not just between adjacent years but across temporally separated observations. Together, the two panels establish that the measure captures a durable feature of each newspaper's language, not transient noise.

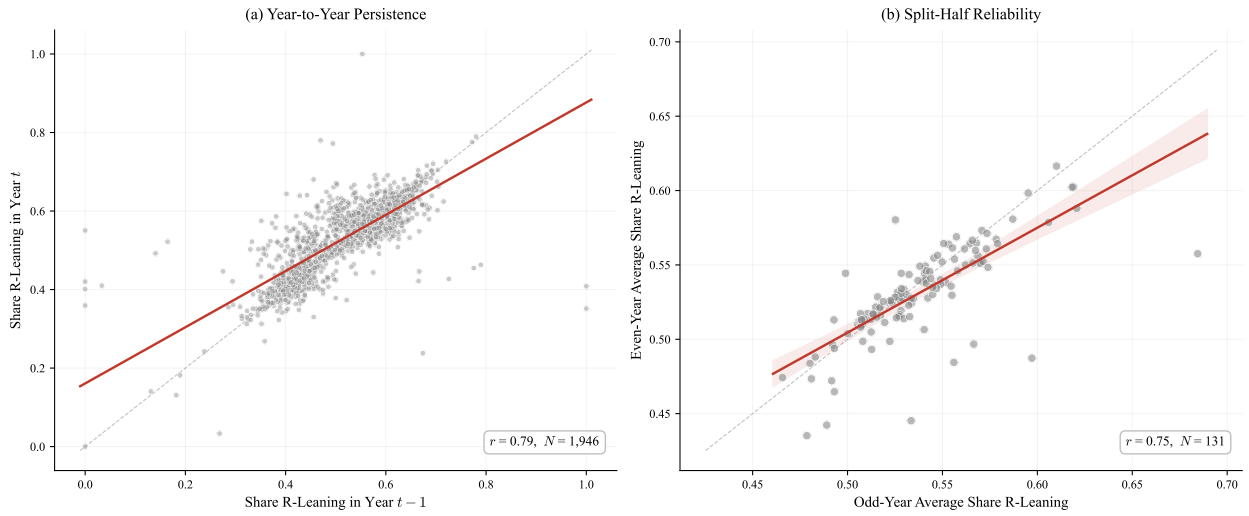

Figure 6: Temporal Stability of the Slant Measure

Notes: Panel (a) plots each newspaper–year's share of R-leaning articles against the same newspaper's value in the preceding year (N = 1,946 newspaper–year pairs). The red line shows the OLS fit; the dashed line marks the 45-degree line. Panel (b) plots each newspaper's average share of R-leaning articles computed over odd years against the average over even years (N = 131 newspapers). The shaded region shows the 95% confidence band. The Pearson correlations are $r = 0.80$ and $r = 0.76$, respectively (both $p < 0.001$).

D.4 External Validation

Convergence and stability establish that the measure is precise and consistent, but not that it captures something politically meaningful. Figure 7 provides external validation: the cross-sectional correlation between a newspaper's average Share R-leaning and the pre-NAFTA Republican presidential vote share in its home county is $r = 0.35$ ($p < 0.001$). Newspapers in Republican-leaning counties do produce more R-leaning text, consistent with the demand-side mechanism documented by Gentzkow and Shapiro (2010). The correlation is moderate

rather than perfect, as expected: vote share reflects many factors beyond media language, and the slant measure captures linguistic—not ideological—positioning.

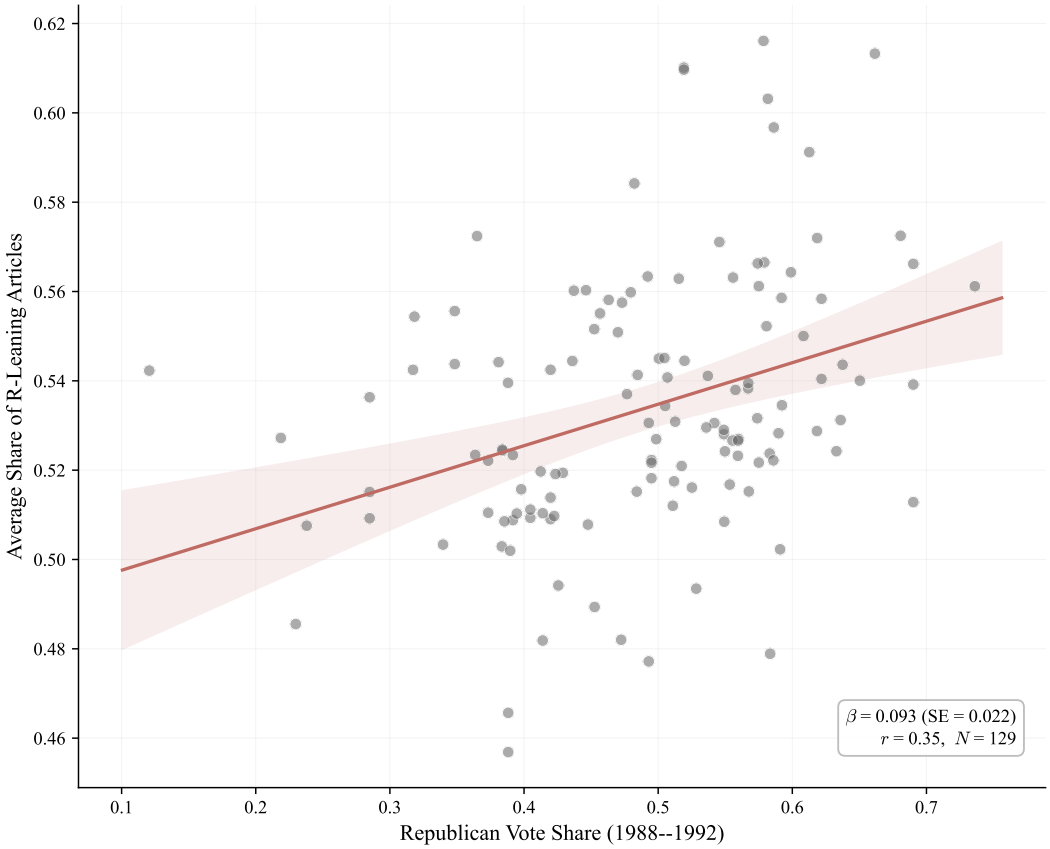

Figure 7: External Validation: Newspaper Slant vs. Local Republican Vote Share

Notes: Each point represents one of 131 estimation-sample newspapers. The x-axis is the Republican two-party vote share averaged over the 1988 and 1992 presidential elections (both pre-NAFTA). The y-axis is the newspaper's average share of Republican-leaning articles (1987–2004). Pearson $r = 0.35$ ($p < 0.001$).

D.5 A Ruler, Not a Scale

The preceding evidence shows that the measure converges, is stable, and correlates with political outcomes. But a natural concern remains: the "zero point" of the measure—the boundary between R-leaning and D-leaning text—is defined by congressional rhetoric, which may not align with how readers or editors perceive partisanship. A newspaper that scores 55% R-leaning need not be perceived as conservative; its language simply resembles

Republican congressional speech more often than Democratic speech.

This concern does not affect the identification strategy. Table 7 decomposes the variance of Share R-leaning: newspaper fixed effects account for 10% and year effects for 65%, leaving 25% as the within-newspaper, within-year residual that the event study identifies. The newspaper fixed effects absorb any time-invariant differences in absolute level—including whatever "miscalibration" exists between the congressional speech boundary and a newspaper's true partisan positioning. The identification requires only that the measure function as a *consistent ruler*: if a trade shock causes a newspaper to shift toward more Republican-associated language, the measure must detect that shift, regardless of where the newspaper started on the absolute scale. The convergence and stability results above confirm that it does.

Table 7: Variance Decomposition of Slant Measures

	Total variance	Newspaper FE	Year FE	Residual
Share R-leaning	0.008	10.0%	65.0%	25.0%

Notes: Decomposition of variance in each outcome variable into the share explained by newspaper fixed effects (cross-sectional), year fixed effects (common time trends), and the residual. The residual is the within-newspaper, within-year variation that identifies the event study coefficients. For both measures, year effects dominate, reflecting common national political dynamics.

E Sample Description

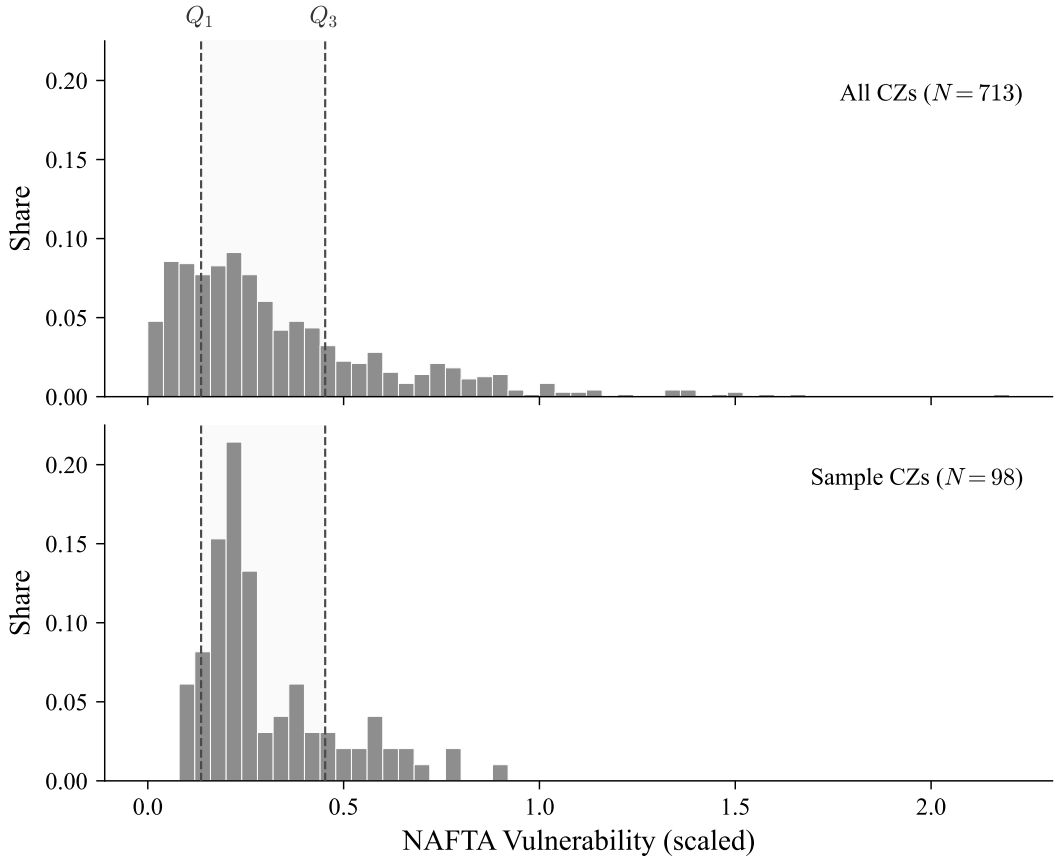

Figure 8: Distribution of NAFTA Vulnerability: All CZs vs. Sample

Notes: Each bar shows the share of commuting zones in a given vulnerability bin, colored using the same sextile bins as Figure 1. The top panel shows all 713 commuting zones; the bottom panel shows the 98 commuting zones in the estimation sample.

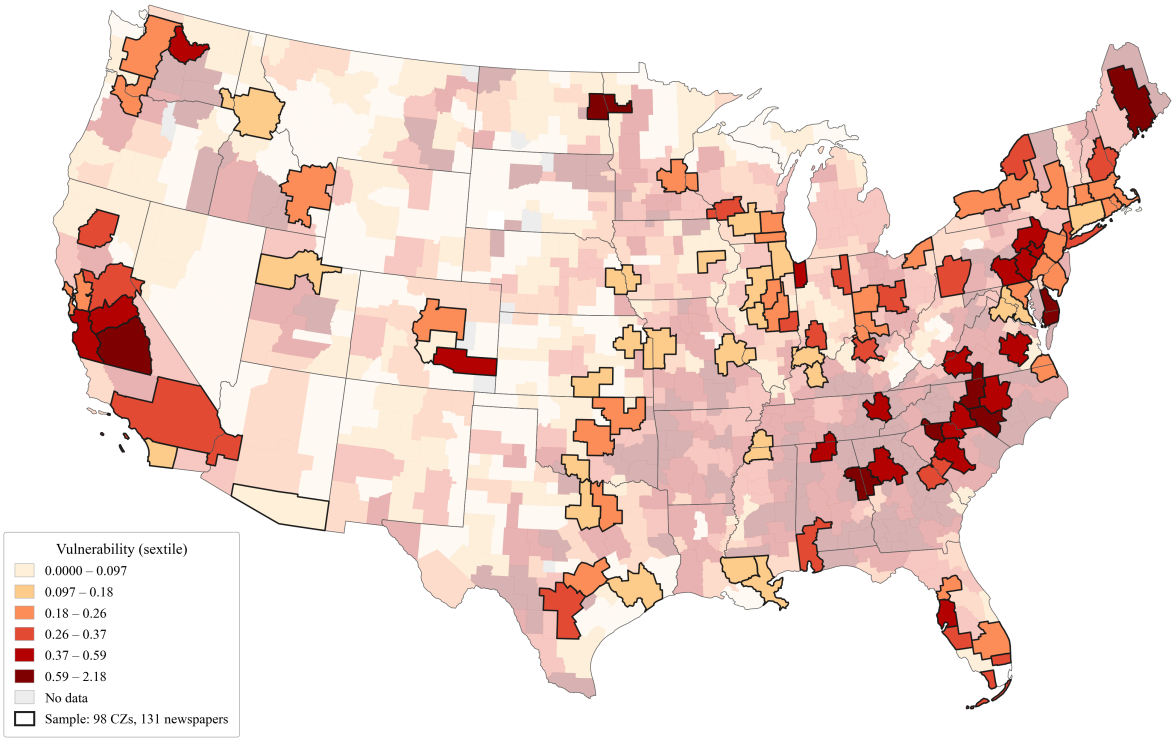

Figure 9: Sample Commuting Zones and NAFTA Vulnerability

Notes: The 98 sample commuting zones (131 newspapers) are shown at full color with dark borders; non-sample areas are faded. The sample spans the full range of the national vulnerability distribution, as shown in Figure 8.

Table 8: Estimation Sample by Year

Year	Newspapers	CZs	Total Articles	Articles/Paper
1987	49	40	1,480,079	30,205
1988	63	52	1,781,346	28,275
1989	81	66	2,395,945	29,579
1990	102	80	2,298,839	22,537
1991	119	90	2,039,856	17,141
1992	131	98	3,534,550	26,981
1993	131	98	3,802,428	29,026
1994	131	98	3,723,611	28,424
1995	131	98	3,439,734	26,257
1996	128	96	3,774,167	29,485
1997	129	97	3,700,828	28,688
1998	128	96	3,639,850	28,436
1999	129	97	3,156,642	24,470
2000	129	97	3,035,071	23,527
2001	127	97	2,488,650	19,595
2002	125	94	2,484,274	19,874
2003	123	94	2,855,957	23,219
2004	121	93	2,973,087	24,570
Mean	115	87	2,922,495	25,571

Notes: The estimation sample is an unbalanced panel of 131 daily newspapers in 98 commuting zones from 1987 to 2004 (2,077 newspaper–year observations; 52.6 million articles). Two national newspapers and four non-daily publications are excluded. The number of newspapers varies across years because some newspapers entered the NewsLibrary digital archive after 1987. Articles/Paper reports the mean number of articles per newspaper in each year.

Table 9: Newspapers in the Estimation Sample

	Newspaper	St	Art/Yr		Newspaper	St	Art/Yr
1	Anniston Star	AL	6,688	34	Atlanta Journal-Constitution	GA	46,551
2	Huntsville Times	AL	14,127	35	Gazette (Cedar Rapids)	IA	27,942
3	Mobile Register	AL	33,984	36	Lewiston Morning Tribune	ID	12,939
4	Arizona Daily Star	AZ	13,447	37	Post Register (Idaho Falls)	ID	7,139
5	Daily Breeze (Torrance)	CA	30,994	38	Chicago Sun-Times	IL	43,258
6	Daily News of Los Angeles	CA	39,317	39	Herald & Review (Decatur)	IL	7,432
7	Fresno Bee	CA	16,886	40	Journal Gazette (Mattoon)	IL	15,198
8	Modesto Bee	CA	10,413	41	Journal Star (Peoria)	IL	33,015
9	Press-Enterprise	CA	24,973	42	Pantagraph (Bloomington)	IL	17,614
10	Press-Telegram (Long Beach)	CA	25,098	43	State Journal-Register	IL	11,951
11	Redding Record Searchlight	CA	6,733	44	Times-Courier (Charleston)	IL	11,199
12	Sacramento Bee	CA	18,570	45	Evansville Courier	IN	13,609
13	San Diego Union-Tribune	CA	22,946	46	Herald-Times (Bloomington)	IN	10,849
14	San Jose Mercury News	CA	40,384	47	Journal Gazette (Fort Wayne)	IN	19,642
15	San Francisco Chronicle	CA	31,147	48	News-Sentinel	IN	8,162
16	Orange County Register	CA	28,364	49	Post-Tribune	IN	23,654
17	Gazette (Colorado Springs)	CO	15,882	50	Times (Munster)	IN	18,903
18	Pueblo Chieftain	CO	14,234	51	Journal-World (Lawrence)	KS	8,240
19	Rocky Mountain News	CO	41,140	52	Wichita Eagle	KS	17,340
20	Denver Post	CO	17,537	53	Lexington Herald-Leader	KY	23,379
21	New Haven Register	CT	20,039	54	Owensboro Messenger-Inquirer	KY	7,420
22	Hartford Courant	CT	45,727	55	Kentucky Post	KY	11,260
23	Washington Times	DC	28,158	56	Advocate (Baton Rouge)	LA	27,093
24	Delaware State News (Dover)	DE	5,241	57	Times-Picayune	LA	38,202
25	Bradenton Herald	FL	8,736	58	Boston Herald	MA	27,690
26	Fort Pierce Tribune	FL	8,610	59	Sunday Republican (Springfield)	MA	3,549
27	Miami Herald	FL	59,245	60	Union-News (Springfield)	MA	21,874
28	Nuevo Herald	FL	22,587	61	Worcester Telegram & Gazette	MA	27,110
29	Ocala Star-Banner	FL	7,543	62	Capital (Annapolis)	MD	10,947
30	St. Petersburg Times	FL	60,558	63	Sun (Baltimore)	MD	29,992
31	Palm Beach Post	FL	30,551	64	Bangor Daily News	ME	20,755
32	Tampa Tribune	FL	40,338	65	St. Paul Pioneer Press	MN	25,596
33	Augusta Chronicle	GA	39,018	66	Star Tribune (Twin Cities)	MN	15,900

Notes: The 131 daily newspapers in the estimation sample, sorted alphabetically by state. Art/Yr is the mean number of articles per year over each newspaper's sample period. The newspapers are distributed across 98 commuting zones. *Continued on next page.*

Table 10: Newspapers in the Estimation Sample (continued)

	Newspaper	St	Art/Yr		Newspaper	St	Art/Yr
67	St. Louis Post-Dispatch	MO	58,049	100	Patriot-News	PA	27,242
68	Kansas City Star	MO	49,079	101	Philadelphia Daily News	PA	14,581
69	Charlotte Observer	NC	51,725	102	Philadelphia Inquirer	PA	29,758
70	Fayetteville Observer	NC	14,018	103	Pittsburgh Post-Gazette	PA	54,372
71	Greensboro News & Record	NC	29,730	104	Sunday News (Lancaster)	PA	5,559
72	News & Observer	NC	18,459	105	Times Leader	PA	8,672
73	Grand Forks Herald	ND	13,646	106	York Daily Record	PA	11,053
74	Omaha World-Herald	NE	29,707	107	Providence Journal	RI	43,909
75	New Hampshire Union Leader	NH	10,334	108	Herald-Journal (Spartanburg)	SC	10,828
76	Hunterdon County Democrat	NJ	9,014	109	State (Columbia)	SC	17,663
77	Press of Atlantic City	NJ	27,386	110	Knoxville News-Sentinel	TN	16,042
78	Star-Ledger	NJ	63,334	111	Commercial Appeal	TN	37,718
79	Record (New Jersey)	NJ	24,413	112	Austin American-Statesman	TX	35,950
80	Post-Standard	NY	33,804	113	Fort Worth Star-Telegram	TX	44,658
81	Staten Island Advance	NY	17,380	114	Houston Chronicle	TX	60,670
82	Syracuse Herald American	NY	7,235	115	San Antonio Express-News	TX	25,316
83	Syracuse Herald-Journal	NY	14,243	116	Dallas Morning News	TX	56,558
84	Buffalo News	NY	43,717	117	Deseret News	UT	55,798
85	Times Union	NY	32,046	118	Salt Lake Tribune	UT	26,971
86	Watertown Daily Times	NY	26,952	119	Richmond Times-Dispatch	VA	33,497
87	Akron Beacon Journal	OH	33,189	120	Roanoke Times	VA	24,874
88	Columbus Dispatch	OH	28,233	121	Virginian-Pilot	VA	25,038
89	Dayton Daily News	OH	26,210	122	Seattle Post-Intelligencer	WA	23,793
90	Plain Dealer	OH	36,576	123	Seattle Times	WA	43,883
91	Cincinnati Post	OH	34,419	124	News Tribune	WA	10,792
92	Lawton Constitution	OK	11,798	125	Wenatchee World	WA	9,318
93	Daily Oklahoman	OK	32,947	126	Capital Times	WI	15,744
94	Tulsa World	OK	41,861	127	Journal Times	WI	12,925
95	Oregonian	OR	24,131	128	La Crosse Tribune	WI	17,490
96	Daily Pennsylvanian	PA	1,994	129	Milwaukee Sentinel	WI	11,409
97	Intelligencer Journal (Lancaster)	PA	16,857	130	Milwaukee Journal	WI	18,768
98	Lancaster New Era	PA	14,420	131	Wisconsin State Journal	WI	15,965
99	Morning Call	PA	43,747				

F Model Diagnostics and Partisan Vocabulary

Table 11 reports accuracy, regularization, and feature-selection statistics for each Congress window. Table 12 lists the 15 most influential features averaged across all windows. Tables 13–15 show the top 10 partisan phrases for each of the nine Congress windows individually, documenting how the partisan vocabulary evolves over time.

Table 11: LASSO Model Diagnostics by Congress

Congress	Training speeches		Train accuracy	Selected features		Partisan gap
	Rep.	Dem.		$\hat{\beta} > 0$	$\hat{\beta} < 0$	Δ_t
100	39,908	42,476	0.684	2,321	2,444	1.094
101	35,238	42,468	0.692	2,294	2,335	1.076
102	36,364	44,317	0.692	1,670	1,794	1.016
103	37,675	38,751	0.712	2,403	2,487	1.409
104	55,108	48,837	0.701	2,154	1,961	1.266
105	39,160	32,671	0.700	1,695	1,544	1.148
106	38,792	33,438	0.711	1,540	1,550	1.308
107	26,937	26,379	0.735	1,956	1,838	1.666
108	29,861	30,650	0.757	2,042	2,022	2.092

Notes: Each row corresponds to a single-Congress rolling window. Training speeches are all Republican and Democratic floor speeches in that Congress. Train accuracy is the in-sample accuracy of the cross-validated LASSO model (5-fold CV). Selected features are the number of non-zero LASSO coefficients (positive = Republican-associated, negative = Democrat-associated), where features include both unigrams and bigrams restricted to the shared congressional–newspaper vocabulary. The partisan gap $\Delta_t = \mu_{R,t} - \mu_{D,t}$ is the difference in mean net slant between all Republican and all Democratic legislators, used for normalization. The training accuracy of 68–76% reflects genuine partisan signal discrimination while training on all legislators, including moderates.

Table 12: Top Partisan Features Identified by LASSO

	Republican-associated				*Democratic-associated*		
	Feature	$\bar{\beta}$	N_t		Feature	$\lvert\bar{\beta}\rvert$	N_t
1	unborn	53.2	8	1	wealthiest	101.7	8
2	liberal	50.8	9	2	tax break	63.2	8
3	communist	48.5	9	3	Head Start	34.3	9
4	illegal alien	47.8	7	4	richest	33.0	8
5	bureaucracy	45.6	9	5	wealthy	32.3	9
6	bureaucrat	44.1	9	6	oil company	29.6	5
7	big government	37.5	6	7	tax cut	28.3	8
8	raise taxes	33.6	9	8	Pentagon	27.4	9
9	tax increase	27.4	9	9	insurance co.	25.3	6
10	Christian	22.8	8	10	gay	22.0	8
11	businessman	17.8	6	11	worker	18.7	9
12	spending	17.4	9	12	black	17.9	9
13	marriage	16.2	7	13	profit	17.8	9
14	freedom	16.0	9	14	racial	17.7	6
15	violent crime	17.0	5	15	women	12.6	9

Notes: Features are stemmed unigrams or bigrams; unstemmed forms are shown for readability. $\bar{\beta}$ is the average LASSO coefficient across all Congress windows in which the feature is nonzero. N_t is the number of Congress windows (out of 9) in which the feature receives a nonzero coefficient. Features are ranked by $|\bar{\beta}|$ among those appearing in at least 5 of 9 windows.

Table 13: Top Partisan Phrases by Congress Window (Congresses 100–102)

	100 (1987–88)		101 (1989–90)		102 (1991–92)	
Panel A: Republican phrases ($\hat{\beta} > 0$)						
1	medicar trust	103.4	drown	100.2	democrat major	126.3
2	businessmen	85.7	just plain	98.9	democrat leadership	112.1
3	bring measur	82.8	deficit spend	91.9	unborn	77.5
4	tenet	76.4	democrat leadership	80.9	burdensom	73.4
5	free world	76.0	london	78.2	control spend	64.0
6	republican confer	74.4	beltway	75.8	privat properti	62.4
7	welfar program	70.3	afghanistan	72.6	increas spend	55.1
8	liber democrat	67.8	sham	72.0	communist	53.1
9	tax american	65.4	sequest	71.7	rais tax	53.0
10	democrat leadership	64.2	businessmen	68.5	feder spend	51.1
Panel B: Democratic phrases ($\hat{\beta} < 0$)						
1	subcommitte chair	−199.1	customari	−190.6	customari	−147.0
2	arm race	−155.2	wealthiest	−129.4	congression black	−113.0
3	congression black	−126.8	militari budget	−124.5	militari budget	−69.1
4	star war	−121.8	subcommitte chair	−105.8	wealthiest	−65.6
5	head start	−111.4	wealthi	−103.8	work famili	−61.0
6	poorest	−101.0	bail	−78.6	educ health	−57.0
7	right american	−87.8	presid say	−77.2	digniti	−56.5
8	black histori	−85.6	democrat altern	−76.1	squeez	−56.4
9	militari budget	−84.1	homeland	−74.7	hope presid	−52.5
10	auto	−76.6	labor standard	−74.5	famili valu	−52.3

Notes: The 10 features with the largest positive (Panel A) and most negative (Panel B) LASSO coefficients for each Congress. Features are Porter-stemmed unigrams and bigrams restricted to the shared congressional–newspaper vocabulary. Coefficients ($\times 10^3$) are from a cross-validated L1 logistic regression trained on all Republican and Democratic speeches.

Table 14: Top Partisan Phrases by Congress Window (Congresses 103–105)

	103 (1993–94)		104 (1995–96)		105 (1997–98)	
Panel A: Republican phrases ($\hat{\beta} > 0$)						
1	higher tax	122.0	preserv protect	97.2	death tax	170.9
2	largest tax	112.7	largest tax	96.5	lower tax	117.7
3	democrat leadership	106.8	welfar state	90.3	big govern	108.9
4	govern control	103.9	clinton	85.4	accordingli	104.7
5	govern regul	97.8	liber democrat	83.5	bureaucraci	81.3
6	hike	97.2	big govern	72.6	obstruct	80.8
7	term limit	93.9	freshman class	70.3	save medicar	71.8
8	big govern	82.1	communist	65.7	liber	68.3
9	send troop	81.2	promis american	62.7	govern program	68.1
10	spender	76.2	spend control	51.1	bureaucrat	67.5
Panel B: Democratic phrases ($\hat{\beta} < 0$)						
1	wealthiest	−185.4	republican colleagu	−237.6	beam	−184.6
2	increas defens	−140.2	wealthiest	−143.5	republican colleagu	−123.7
3	futur children	−115.9	congression black	−116.5	wealthiest	−117.3
4	work famili	−107.8	million children	−112.4	republican major	−91.8
5	tax break	−105.8	republican leadership	−87.1	tax break	−84.4
6	congression black	−104.0	cut educ	−75.3	republican leadership	−81.0
7	tax benefit	−87.8	senior pay	−75.1	democrat altern	−76.5
8	earn incom	−85.1	speaker republican	−65.4	pell	−71.4
9	discretionari spend	−84.2	presid clinton	−64.7	school construct	−70.0
10	worker right	−81.7	affirm action	−63.6	class size	−68.0

Notes: See Table 13. Congress 103 spans the NAFTA implementation year (1994). Congress 104 coincides with the Republican Revolution.

Table 15: Top Partisan Phrases by Congress Window (Congresses 106–108)

	106 (1999–00)		107 (2001–02)		108 (2003–04)	
Panel A: Republican phrases ($\hat{\beta} > 0$)						
1	bureaucraci	130.8	accordingli	156.9	illeg alien	218.4
2	accordingli	121.7	member urg	139.3	support rule	172.0
3	death tax	105.2	bureaucrat	83.8	death tax	121.4
4	democrat control	83.4	burdensom	80.8	presid georg	116.9
5	tax burden	82.4	titl unit	76.8	liber	99.6
6	raid	79.0	geneva	75.2	lower tax	97.4
7	member republican	69.8	republican member	75.1	mankind	93.4
8	titl unit	65.4	liber	75.0	bureaucraci	91.6
9	xxi	62.5	standard live	74.6	dictatorship	91.0
10	member urg	61.9	chairmen	74.4	republican confer	90.1
Panel B: Democratic phrases ($\hat{\beta} < 0$)						
1	beam	−230.7	livabl	−198.8	fall short	−134.8
2	livabl	−136.5	black caucu	−184.9	hypocrisi	−124.8
3	wealthiest	−120.3	beam	−171.4	underfund	−117.8
4	republican colleagu	−120.1	republican colleagu	−151.7	republican leadership	−110.4
5	tax break	−112.2	republican leadership	−144.4	insur industri	−110.3
6	republican leadership	−106.1	class size	−126.3	make million	−109.8
7	nonprolifer	−88.8	unilater	−121.2	privat insur	−105.8
8	democrat altern	−87.8	latino	−114.5	republican colleagu	−102.7
9	trade deficit	−85.8	martin king	−103.4	presid say	−94.8
10	richest	−85.8	technic assist	−97.1	congression black	−89.3

Notes: See Table 13.

G Robustness

This appendix collects robustness checks referenced in Section 6.4. Table 16 addresses panel balance. Figures 10–12 examine classification threshold sensitivity. Tables 17–19 document that trade-specific language does not drive the results. Figure 13 reports randomization inference.

Table 16: Balanced Panel Robustness

	(1) Full sample 131 papers, 98 CZs	(2) Balanced 46 46 papers, 39 CZs	(3) 1992–2004 131 papers, 98 CZs
Share R-leaning	0.096***	0.115***	0.077**
	(0.032)	(0.030)	(0.031)
Paper FE	Yes	Yes	Yes
Division × Year FE	Yes	Yes	Yes
China shock × Year	Yes	Yes	Yes
Manuf. share × Year	Yes	Yes	Yes
Observations	2,077	828	1,663

Notes: Column (1) reproduces the baseline DiD from column (3) of Table 3. Column (2) restricts the sample to the 46 newspapers present in all 18 years (1987–2004). Column (3) restricts the sample period to 1992–2004, when all matched newspapers are present. All specifications include the full set of fixed effects and controls from column (3) of Table 3. Standard errors clustered at the commuting zone level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

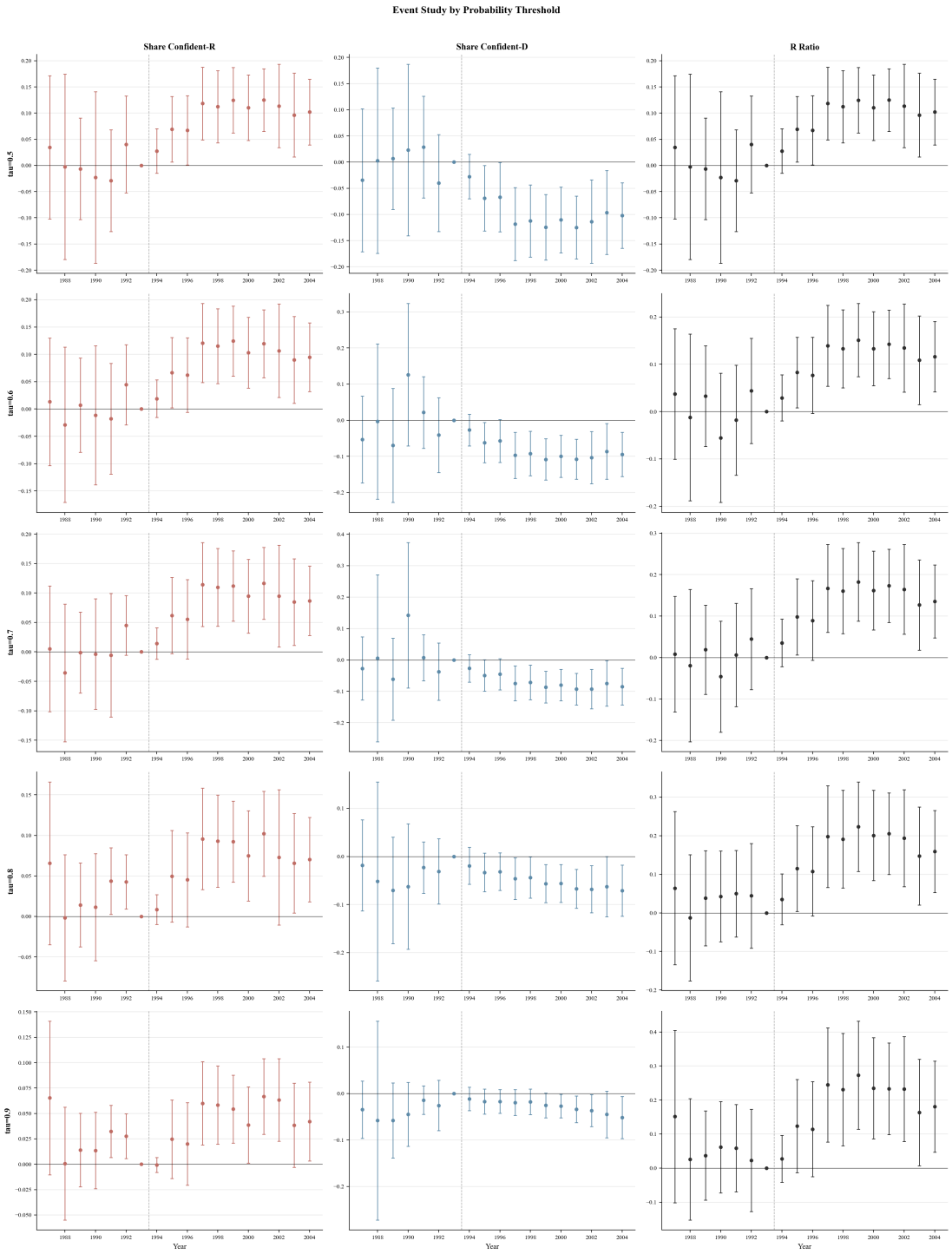

Figure 10: Probability-Threshold Robustness

Notes: Rows correspond to probability thresholds $\tau = 0.5, 0.6, 0.7, 0.8, 0.9$. Columns show Share Confident-R, Share Confident-D, and R Ratio $= R/(R + D)$. All specifications include the full set of fixed effects and controls. Vertical bars indicate 95% confidence intervals clustered at the commuting zone level.

Figure 11 superimposes the Share R-leaning event study across all five probability thresh-

olds. Higher thresholds attenuate the level of the coefficients—fewer articles qualify as "confident R"—but the post-NAFTA pattern is preserved at all cutoffs.

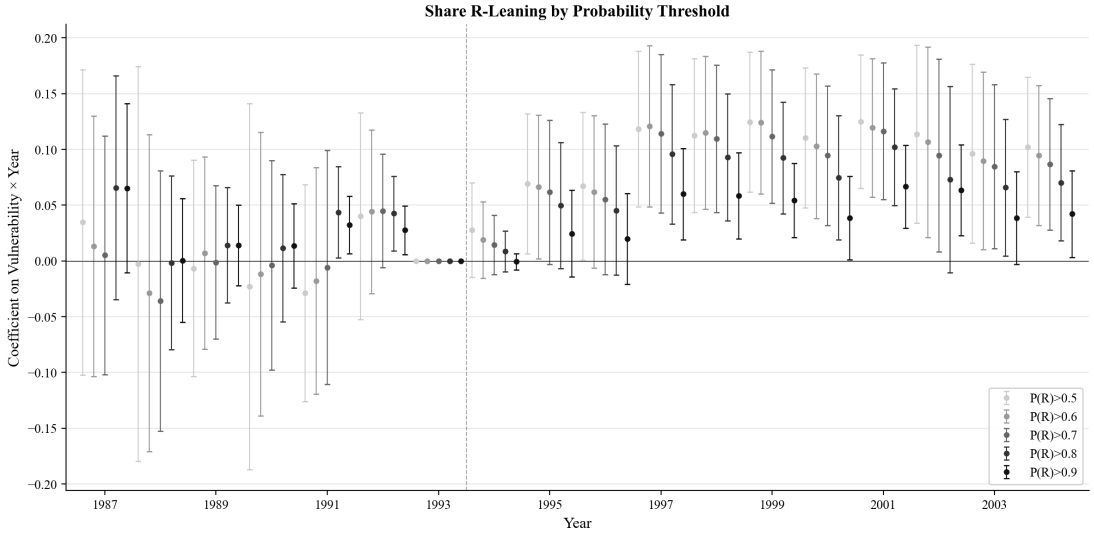

Figure 11: Share R-Leaning at Different Probability Thresholds

Notes: Each series plots event study coefficients for the share of articles classified as R-leaning at the indicated probability threshold τ. Darker colors correspond to stricter thresholds. All specifications include newspaper, year, and division-by-year fixed effects plus China shock and manufacturing share controls interacted with year.

Figure 12 plots the R Ratio at each threshold. The post-NAFTA effect increases monotonically with τ, from 0.095 at $\tau = 0.5$ to 0.188 at $\tau = 0.9$, confirming that the rightward shift is concentrated among articles with clearly identifiable partisan language.

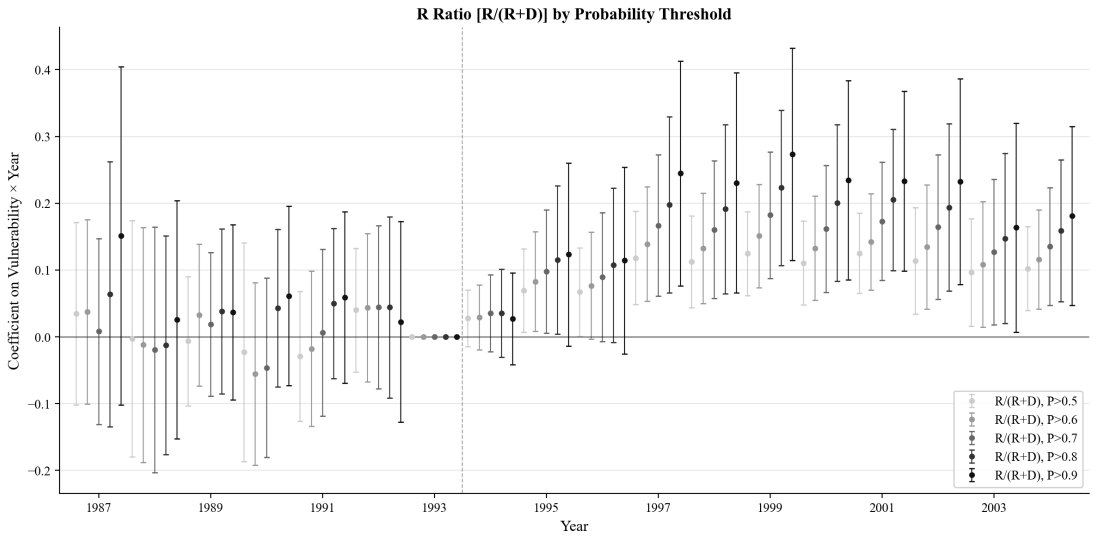

Figure 12: R Ratio at Different Probability Thresholds

Notes: The R Ratio is defined as $R/(R+D)$ among articles exceeding the probability threshold in either direction. The post-NAFTA mean effect increases from 0.095 at $\tau = 0.5$ to 0.188 at $\tau = 0.9$, indicating that the rightward shift is concentrated in clearly partisan articles.

The following tables verify that trade-specific language does not drive the main results. Table 17 decomposes the partisan vocabulary by topic, showing that trade features account for only 0.3% of the total LASSO coefficient weight. Table 18 zeros all trade-related coefficients and re-estimates the DiD, confirming negligible change. Table 19 estimates the DiD separately for each topic category.

Table 17: Topic Decomposition of the Partisan Vocabulary

Topic	Features	Active	% of Weight
Government & procedure	350	332	5.4
Tax & fiscal	238	216	4.4
Defense & foreign	131	121	1.7
Social policy	131	109	1.6
Healthcare	106	96	1.4
Education	63	55	0.6
Crime & justice	43	41	0.6
Civil rights & identity	33	31	0.6
Environment & energy	33	33	0.5
Trade	**25**	**24**	**0.3**
Other (general partisan)	7,074	6,709	83.0

Notes: Each row reports a topic category's share of the total absolute LASSO coefficient weight, averaged across all nine Congress windows. "Features" is the number of unique features assigned to each category; "Active" is the number with nonzero coefficients in at least one window. Topics are defined by keyword classification of stemmed features. Trade features include *trade*, *tariff*, *nafta*, *export*, *gatt*, *wto*, *quota*, and their bigrams (25 features total). The "Other" category contains general-purpose partisan terms not assigned to any substantive topic.

Table 18: Trade-Feature Zeroing: DiD Estimates with Trade Coefficients Set to Zero

		Narrow (26 features)		Broad (81 features)	
Outcome	Baseline	Coef	% Δ	Coef	% Δ
Share R-Leaning	0.096	0.096	+0.5	0.096	+0.3

Notes: Each column reports the DiD coefficient on $Vul_c \times Post_t$ from the fully controlled specification (newspaper, year, and division-by-year FE; China shock and manufacturing share controls). "Baseline" uses the original LASSO coefficients. "Narrow" zeros 26 trade-specific features (0.4% of total $|\beta|$ weight); "Broad" zeros 81 features including all forms of *import* (0.9% of weight). Articles are re-scored from scratch after zeroing, then re-aggregated to the newspaper–year panel before re-estimating the DiD. All percentage changes are less than 1% for the main outcomes.

Table 19: Share of R-Leaning Articles by Topic: DiD Estimates

Topic	% of Articles	Coef	SE	p-value
All articles	100.0	0.0956	0.0318	0.0034
Education	18.8	0.1164	0.0378	0.0027
No specific topic	30.4	0.1020	0.0305	0.0012
Defense & foreign	13.5	0.0832	0.0292	0.0054
Trade	**2.9**	**0.0753**	**0.0361**	**0.0393**
Healthcare	10.7	0.0647	0.0443	0.1475
Social policy	16.8	0.0617	0.0303	0.0443
Environment & energy	3.2	0.0611	0.0411	0.1405
Gov't & procedure	22.2	0.0596	0.0293	0.0449
Civil rights & identity	8.0	0.0487	0.0302	0.1104
Crime & justice	11.8	0.0446	0.0289	0.1266
Tax & fiscal	6.7	−0.0029	0.0412	0.9446

Notes: Each row reports the DiD coefficient on $Vul_c \times Post_t$ with the topic-specific share of R-leaning articles as the dependent variable. An article is assigned to a topic if any of the topic's keyword features have a nonzero value in its feature vector; articles can belong to multiple topics. "No specific topic" denotes articles that do not match any of the ten keyword categories. All specifications include newspaper, year, and division-by-year FE, plus China shock and manufacturing share controls interacted with year. Standard errors clustered at the CZ level.

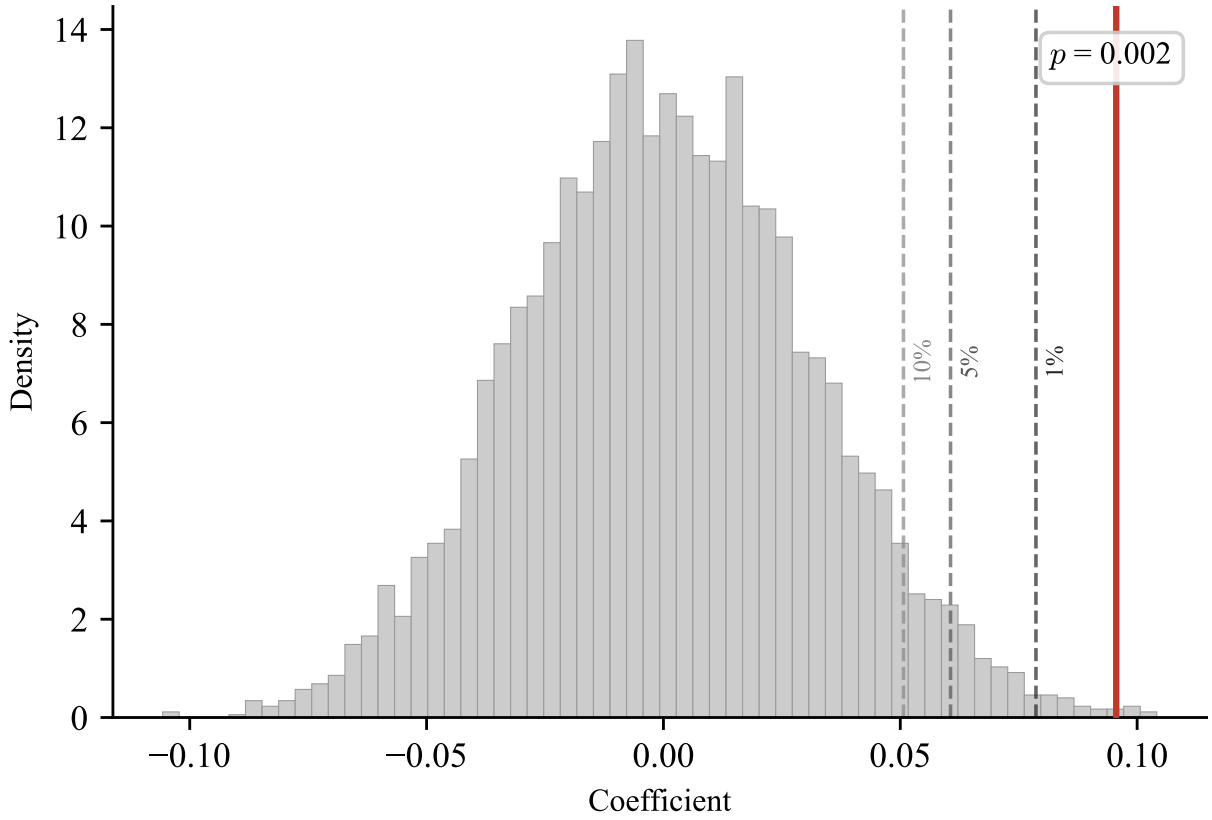

Figure 13: Permutation Distribution of DiD Coefficients (5,000 CZ-Level Permutations)

Notes: The figure shows the distribution of DiD coefficients from 5,000 random permutations of the CZ-level vulnerability assignment. The solid red line marks the actual estimate. The permutation p-value is two-sided. The permutations preserve the panel structure within each CZ and use the Frisch–Waugh–Lovell theorem for computational efficiency. The specification includes newspaper, year, and division-by-year fixed effects plus China shock and manufacturing share controls interacted with year.

H Alternative Training Sample: Partisan Core

The baseline specification trains the slant model on *all* Republican and Democratic speeches. As a robustness check, I restrict the training sample to the most partisan 20% of legislators in each Congress, identified by their Nokken–Poole first-dimension scores (Lewis et al., 2024). Nokken–Poole scores are session-specific ideal-point estimates derived from roll-call voting records; unlike fixed DW-NOMINATE scores, they allow a legislator's estimated ideology to

shift across Congresses. For each Congress, I classify legislators in the top 20th percentile of the Nokken–Poole dimension-1 distribution as the Republican core and those in the bottom 20th percentile as the Democratic core.

Table 20 summarizes the core training sample. In each Congress, approximately 87 legislators (out of ∼435 House members) are classified as core partisans, contributing 13,000–23,000 speeches. Core Republicans have mean Nokken–Poole scores of 0.56–0.66, while core Democrats average −0.55 to −0.59—substantially more extreme than the full-party means (0.34–0.43 for all Republicans, −0.31 to −0.38 for all Democrats). Training on this subset should yield a sharper partisan vocabulary, at the cost of a smaller and potentially less representative training sample.

Table 20: Core Partisan Training Sample by Congress

	Legislators		Speeches		Mean NP Dim1	
Congress	Core R	Core D	Core R	Core D	Core R	Core D
100	35	52	13,066	10,397	0.56	−0.55
101	35	52	8,668	9,177	0.58	−0.57
102	34	54	11,866	10,861	0.58	−0.57
103	35	51	13,473	8,450	0.59	−0.59
104	48	41	9,653	11,377	0.65	−0.57
105	47	43	8,031	9,412	0.64	−0.57
106	45	42	8,328	8,417	0.66	−0.58
107	47	42	5,454	7,697	0.66	−0.58
108	47	40	6,697	8,772	0.65	−0.57

Notes: Core Republicans (Democrats) are House members whose Nokken–Poole first-dimension score falls in the top (bottom) 20th percentile of the chamber-wide distribution in each Congress. NP Dim1 values are mean scores within each core group.

Table 21 reports the DiD results using the core-trained slant measure. All specifications and controls mirror those in Table 3. The pooled estimate for Share R-leaning is 9.8 percentage points ($p < 0.01$)—nearly identical to the baseline estimate of 9.6 percentage points. The short-term (1994–1998) and long-term (1999–2004) decomposition also matches the gradual build-up pattern observed in the baseline.

Table 21: DiD Estimates: Core 20% Training Sample

	(1)	(2)	(3)	(4)	(5)
		Pooled		*Short-term*	*Long-term*
Share R-leaning	0.102***	0.097***	0.098***	0.084***	0.110***
	(0.024)	(0.029)	(0.029)	(0.030)	(0.032)
Paper FE	Yes	Yes	Yes	Yes	Yes
Division × Year FE	Yes	Yes	Yes	Yes	Yes
Manuf. share × Year		Yes	Yes	Yes	Yes
China shock × Year			Yes	Yes	Yes
Observations	2,077	2,077	2,077	2,077	

Notes: This table replicates Table 3 using the slant model trained on the most partisan 20% of legislators (by Nokken–Poole dimension-1 scores) rather than all Republican and Democratic speeches. All other pipeline settings—shared vocabulary filter, 5-fold cross-validation, unigram+bigram features—are identical to the baseline. Standard errors clustered at the commuting zone level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The stability across training samples is expected. The Share R outcome depends only on whether an article's predicted probability exceeds 0.5, not on the magnitude of the score. Training on core legislators sharpens the coefficient vector—each selected feature receives a larger absolute weight—but this shifts the entire score distribution without changing which articles cross the 0.5 threshold. The near-identical estimates confirm that the compositional shift is robust to the choice of training sample.